Preprint

Phraqtal Field

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Abstract.

This work develops the *Phraqtal Field* as the organisational framework of the Universal Quantum Foam Hypothesis (UQSH). Its starting point is the Qu-Foundation Q, whose physically readable field form Φ acts in different regimes as free radiation, bound matter, gravitation, saturation, release and observability. The Phraqtal Field does not describe these phenomena as separate substances, but as different organisational states of a continuous field medium.

The designations *Qu-Field* and *Qu-Foam* used in earlier works are not continued here as the central field designation. They are replaced by the more precise term *Phraqtal Field*, which clearly distinguishes the physically observable, regime-dependent and scale-nested organisational form of the field medium from other fields and existing field theories.

The central focus is a scale-relative assignment between atomic and galactic organisational roles. Atomic nuclei and galactic central regions take on the role of central binding and saturation organisations. Electrons, gas, stellar populations, satellites and outer field structures each form the outer bound occupancy. Hydrogen and helium are classified as the first open and closed basic families of matter, respectively. Heavy excited or unstable atomic states form the equivalent of highly saturated galactic central organisations such as quasars.

The work connects the empirically determined bimodality of galactic rotation profiles with the dynamic Q-switch introduced as a theoretical organisational quantity. The shape axis q_{B} describes the spatial structure of the outer field response, while the activation axis q_{A} describes the switching state between inner binding and outer reorganisation. This results in a multidimensional classification of galaxies according to binding, openness, coherence, saturation, visibility and reorganisation capacity.

Quasars are interpreted as cyclic saturation and release centres. Their high-energy radiation, the organisation of minimally coupling-capable forms of matter, outer gas and star-formation zones, and the feedback to the host galaxy form a common reorganisation cycle. Visibility is understood as a regime-dependent coupling event: physical field organisation and immediate observability do not necessarily coincide.

The Phraqtal Field thus provides a testable organisational vocabulary for the joint analysis of acceleration profiles, baryonic structure, spectra, star formation, satellite coupling and quasar activity.

Keywords: Phraqtal Field; Universal Quantum Foam Hypothesis; Qu-Foundation; matter for-

mation; visibility; galactic rotation; bimodality; Q-switch; quasars; atom–galaxy organisation

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1. Introduction and Position of the Phraqtal Field

1.1. Introduction

The physical description of matter, radiation, gravitation and observation usually uses different theoretical languages. Particles, fields, spacetime, energy, spectra and galactic dynamics appear as separate starting categories. The Universal Quantum Foam Hypothesis (UQSH) takes a different approach: it treats these phenomena as different states and readable forms of a continuous field medium [9].

The starting point of this medium is the Qu-Foundation Q. It does not denote an additional substance in space or a classical ether. Q denotes the deepest foundation, which is not directly measurable itself, from which a physically real field form Φ emerges. What is measurable is not Q itself, but its regime-dependent effects: binding, field motion, radiation, gravitation, saturation, release, matter and visibility.

$$Q \longrightarrow \Phi \longrightarrow B(\Phi).$$

$B(\Phi)$ denotes the moving and organised field form in which free radiation, bound matter and gravitational field responses take shape. Within this framework, light is the freely propagating form of local field reorganisation. Matter is its bound and coherently stabilised form. Gravitation describes the regime-effective binding, tension and curvature side of the same organisation.

This work introduces the term *Phraqtal Field* for this common mode of organisation. The designations *Qu-Field* and *Qu-Foam* used in earlier works are not adopted as the central field designation. They remain historical names from the original UQSH working context, but do not yet describe the physical field reality with sufficient precision.

The term *Phraqtal Field*, by contrast, denotes the physically real, coupling-capable and regime-dependently readable organisational form of the field medium. It distinguishes this field reality, linguistically and conceptually, from established field terms and field theories. The Phraqtal Field is neither an additional classical field alongside known physical fields nor merely a mathematical construction. It denotes the scale-nested mode of organisation in which the different physical states take shape as binding, motion, radiation, matter, gravitation, saturation and reorganisation.

The prefix *Ph* refers to the physical and holographically readable appearance of the field; the lowercase *q* refers to the conceptual connection with the Qu-Foundation, which is not directly measurable. The term *Phraqtal Field* therefore does not replace the UQSH as the overarching hypothesis framework, but specifies the designation of the physically effective field medium described within it. The term does not describe a virtual world or a mere mapping relation. It describes how bound structures on different scales carry common organisational roles and each form a scale-relative readability of the continuous field foundation.

These organisational roles become especially clear in the comparison between atoms and galaxies. Atomic nuclei and galactic central regions take on the role of central binding. Electrons, gas, stars, satellites and outer field structures take on the role of outer bound

organisation. In both cases, coherence, saturation, coupling capacity and reorganisation readiness determine which states remain stably integrated, become visibly activated or are transferred into new forms of binding.

The comparison is made more concrete by two earlier findings on galactic acceleration organisation. First, rotation profiles show a bimodal shape structure with concentrated peak and broadly distributed diffuse outer organisations. Second, a transition strength points to a dynamic activation range in which a characteristic transition scale is switched on [7, 8]. These two properties are transferred here into the language of the Phraqtal Field as the shape axis q_{B} and the activation axis q_{A}.

A particular focus is placed on quasars and highly saturated galactic central regions. They are treated as cyclic release organisations: central inflow and compression lead to saturation; saturation leads to coherent high-energy release; this release acts back on outer gas and matter-formation zones; part of the reorganised matter is then integrated back into the host organisation or returned to the central region.

The work therefore pursues four goals:

1. to define the concept and organisational logic of the Phraqtal Field,
2. to formulate an atom–galaxy assignment of binding, outer organisation and saturation,
3. to classify bimodality and Q-activation as galactic differentiation axes,
4. to establish an empirical testing programme for profiles, matter, visibility and quasar cycles.

The following sections first develop the Qu-Foundation and the regime-dependent readability of the Phraqtal Field. Matter formation, element families, satellite coupling, quasar cycles and the atom–galaxy overall matrix are then worked out. Finally, the operational testing architecture is formulated, with which the organisational assignment can be tested against galactic data.

1.2. Position of the Phraqtal Field within the UQSH Working Context

The Phraqtal Field brings together several lines of work already developed within UQSH in a common organisational framework. It does not replace these works, but connects their respective statements about the Qu-Foundation, light, matter, saturation, galactic acceleration organisation and visibility.

1.2.1 Qu-Foundation, Field Motion and Gravitational Light

The ontological basis of the Phraqtal Field lies in the distinction between the Qu-Foundation Q, its physically real field form Φ and the moving field organisation $B(\Phi)$:

$$Q \longrightarrow \Phi \longrightarrow B(\Phi).$$

This structure was developed within *Gravitational Light* as the basis for light, energy, gravitation, matter, time and observation [9]. The Phraqtal Field adopts this foundation and places the focus on the common organisational logic of atomic and galactic systems.

1.2.2 Galactic Acceleration Excess

The baryon-anchored analysis of galactic acceleration excesses forms the dynamic basis for the galactic side of the Phraqtal Field. The positive acceleration excess is read as a nonlocal and regime-dependent field response that remains connected to baryonic organisation:

$$g_{\mathrm{obs}}(r) = g_{\mathrm{bar}}(r) + \Delta g_{\mathrm{Feld}}(r).$$

The baryon-anchored analysis of galactic acceleration excesses forms the dynamic basis for the galactic side of the Phraqtal Field. This empirical question is linked to the SPARC rotation curves and the observed Radial Acceleration Relation [1, 2]. Within the present framework, the positive acceleration excess is read as a nonlocal and regime-dependent field response that remains connected to baryonic organisation [7]:

The Phraqtal Field uses this field response not only to describe a rotation curve. It places its radial shape within a broader context of binding, outer openness, satellite coupling, visibility and matter reorganisation.

1.2.3 Bimodality and Dynamic Activation

The works on bimodal regime structure and the dynamic activation of a transition scale provide the two galactic differentiation axes of the Phraqtal Field. Bimodality distinguishes concentrated and diffuse outer organisations; the activation analysis describes the switching state between inner and outer field organisation [8].

Within the Phraqtal Field, these quantities are combined as

$$q_{\mathrm{B}} = \text{shape axis}, \qquad q_{\mathrm{A}} = \text{activation axis}$$

.

This distinction is essential. The shape of an outer field response and the activation of a transition scale are different properties. Only their combination allows a differentiated classification of galaxies as open, integrated, locally reorganising or highly activated overall organisations.

1.2.4 Matter Formation and Minimal Field Modes

The works on matter formation and electron–positron modes provide the basis for the atomic side of the Phraqtal Field. They do not treat matter as a collection of finished particles from the outset, but as a sequence of increasing field binding, coherence and coupling capacity.

The sequence used here

$$
\begin{array}{c}
\text{release tension} \\
\Downarrow \\
e^- e^+ \\
\Downarrow \\
\text{quark organisation} \\
\Downarrow \\
\text{nucleons} \\
\Downarrow \\
\text{H, He}
\end{array}
$$

places the first coupling-capable forms of matter within the larger cycle of saturation, release and reorganisation.

The Phraqtal Field extends this line to galactic scales. It examines how minimal binding, open and closed matter families, outer coupling and the central saturation state are repeatedly organised in stars, galaxies and quasars.

1.2.5 Independent Contribution of this Work

The independent contribution of the Phraqtal Field consists of five connections:

1. connection of the Qu-Foundation with scale-relative organisational analysis,
2. assignment of atomic and galactic binding roles,
3. classification of hydrogen and helium as early open and closed basic families,
4. connection of bimodality and Q-activation with element families,
5. description of the quasar as a saturation, release, matter-formation and feedback centre.

The Phraqtal Field therefore takes on a connecting position within the UQSH context. It provides the organisational language in which the previously separate topics of light, matter, visibility, galactic dynamics and saturation can be read as parts of a common recursive field order.

1.3. The Phraqtal Field and the Qu-Foundation

The *Phraqtal Field* denotes the physically real, regime-dependently readable organisational form of the Qu-Foundation Q. Within UQSH, the Qu-Foundation is not directly measurable. What can be measured are its field effects: radiation, matter, gravitation, energy, time, coupling, saturation, release and reorganisation.

The physically effective field form of the Qu-Foundation is denoted by Φ:

$$Q \longrightarrow \Phi \longrightarrow B(\Phi).$$

The physical properties of water, air, heat, rigidity, elasticity, radiation, matter and gravitation are different expressions of the Phraqtal Field itself. The Phraqtal Field is therefore not a

quantity subsequently adjusted by mathematics, but the physical reality from which the observable states of physics arise.

Within this interpretation, the Qu-Foundation Q is not measurable and is not mapped onto Φ through a direct mathematical relation. The chain

$$Q \longrightarrow \Phi \longrightarrow B(\Phi)$$

denotes an ontological and conceptual order: the Qu-Foundation forms the unmeasurable basis, Φ denotes the coupling-capable and observable field form, and $B(\Phi)$ describes its moving and organised states. Mathematics does not describe the Qu-Foundation itself, but places the measurable states of the Phraqtal Field within a framework that we can understand.

The grid size ℓ_n is not a property of the Phraqtal Field itself, but a nested unit of measurement and resolution. For large ℓ_n, large-scale super-regimes and their overall organisation can be described. For small ℓ_n, finer sub-regimes, internal couplings and small-scale reorganisation processes can be resolved. The field form $\Phi(\ell_n)$ captured with a particular grid size is therefore the respective observable coupling level of the Phraqtal Field.

Φ is not a virtual projection or merely an image. It is the physically real field form in which the Qu-Foundation becomes readable for a particular regime as motion, tension, curvature, binding and organisation.

The Phraqtal Field describes the recurring organisational mode of this field form across different scales. The lowercase q in the name refers to the Qu-Foundation without treating it as a separate additional substance. The prefix *Ph* denotes the physically holographic readability of the field: each bound organisation does not carry the entire field in reduced form, but rather a scale-relative, coherently coupled organisational role of the whole.

Matter is bound field motion within the Phraqtal Field. Visibility arises when the motion, coherence, frequency, density, duration and geometry of a field organisation match the coupling filter of an observation regime:

$$\mathcal{V} = \mathcal{V}\left(C,\, \nu,\, \rho,\, \tau,\, \mathcal{G},\, \kappa\right).$$

The observable field organisation is therefore always tied to the chosen resolution scale:

$$\Phi_n = \Phi(\ell_n).$$

Increasing ℓ_n makes higher-level regimes and super-regimes visible, while reducing ℓ_n reveals finer sub-regimes and internal coupling states. A structure can therefore be physically organised without becoming visible in every regime. What an observer perceives is not the complete field reality, but the modulation and coupling layer that can connect to the observer's own bound organisation.

1.4. Invariant Field Motion and Scale-Relative Time Order

The Phraqtal Field combines invariant field motion with a scale-relative organisation of time. The field front itself moves at the speed of light c. This invariant remains preserved across

the different regimes. What changes with scale is not the field motion, but the time that a bound organisation needs for its internal reorganisation, its coupling and its visibility.

For a distance L, the field front has the travel time

$$\tau_c = \frac{L}{c}.$$

This travel time describes the immediate propagation of the field. It must be distinguished from the organisation-specific time τ_{org}. The latter describes the time in which a bound field form stabilises, redistributes energy, enters a new coupling relationship or changes its visible form. Thus, within the Phraqtal Field:

$$\text{Field motion} \neq \text{Organisation time}.$$

An electron, an atom, a biological organism, a star or a galaxy does not consist of field fronts that are each slower or faster. They are organisational forms of the same field with different depths and different spatial extents. Their proper times arise from the density of their bindings, the number of their coupled sub-organisations, their degree of saturation and the range of their feedback loops.

The scale order can therefore be read as the ratio of field travel time to organisation time:

$$\Xi_n = \frac{\tau_{\mathrm{org},n}}{\tau_{c,n}}.$$

A small Ξ_n characterises an organisation whose visible and material response lies close to the immediate field motion. A large Ξ_n characterises an organisation with deep feedback loops, high storage capacity and a long reorganisation period. Galaxies and especially their central regions have a high Ξ_n because of their large binding range and their multi-stage matter, radiation and return processes. Atomic systems, on the other hand, can have a high internal reorganisation density despite their small spatial extent.

The different regimes are not separated from one another in time by this. They are active simultaneously, but process field changes with different organisational durations. A process may already form a stable reorganisation on a lower scale while its higher-level galactic or cosmic integration is still developing. Conversely, a large-scale change in saturation can stimulate local processes without immediately dissolving their internal binding structure.

$$R_{n-1} \longleftrightarrow R_n \longleftrightarrow R_{n+1}$$

Regime coupling therefore does not follow a rigid chain of causes, but a layered simultaneity. Each scale remains tied to invariant field motion and at the same time has its own reorganisation time. The Phraqtal Field therefore describes time as a readable order of coupling durations: light shows the immediate motion of the field; matter stores its bound history; sensation and consciousness form the deepest known forms of integrated temporal feedback.

1.5. Coherent Coupling between the Regimes

The regimes of the Phraqtal Field are not separate layers. They form a nested organisational order in which each level has its own spatial range, its own reorganisation time and its own form of visibility. The coupling between these levels takes place through coherence.

A tension front and its internal modulation propagate together with c. Their local return, twisting and reorganisation, however, take place at the finer temporal resolution of the respective underlying sub-regime. This allows a bound organisation to remain internally active while at the same time being embedded in a larger organisation.

$$R_{n-2} \subset R_{n-1} \subset R_n \subset R_{n+1} \subset R_{n+2}.$$

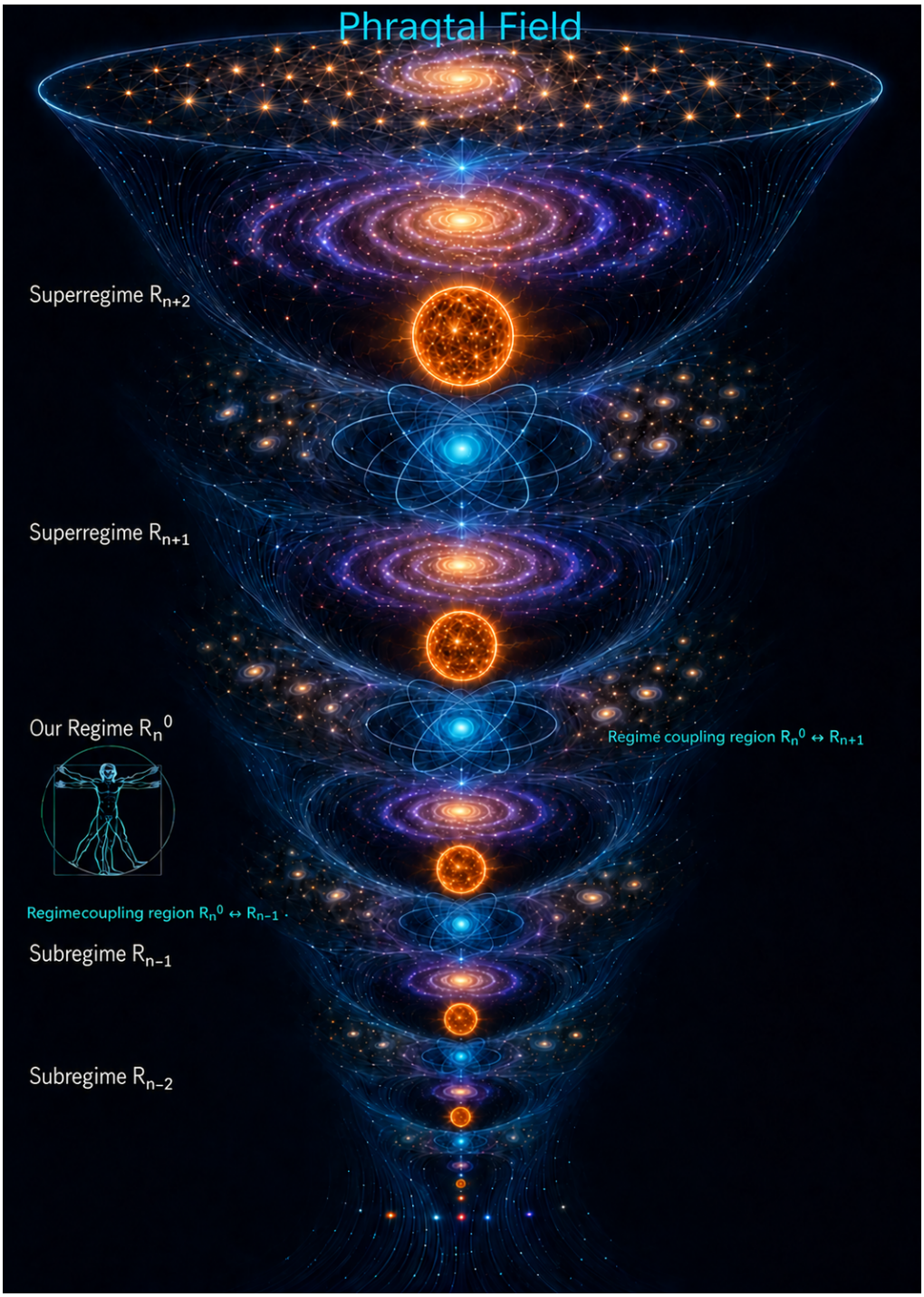


Abbildung 1: Schematic representation of the Phraqtal Field as a nested regime order. The own regime R_n^0 is embedded in underlying sub-regimes and higher-level super-regimes. The transitions between the levels are shown as regime-coupling regions.

The nesting shown in Figure 1 illustrates that the Phraqtal Field does not consist of separate levels. Each regime level has its own resolution, time order and coupling structure, while at the same time remaining embedded in the higher-level and lower-level organisational domains.

A process in R_{n-1} is therefore not understood as a slower copy of a process in R_n. It has its own

internal dynamics, whose feedback is resolved on its own scale. The higher-level organisation specifies the binding space, the large-scale orientation and the load conditions; the lower-level organisation processes these conditions within its own temporal and geometric structure.

In this context, coherence means that the reorganisations of the different scales do not work against one another, but form a shared order that can connect. It can be written as a relation between binding, openness and feedback:

$$C_n = C(K_n, O_n, R_n).$$

Here K_n denotes the binding strength, O_n the organisational openness and R_n the feedback capacity of a structure in regime n. High coherence exists when an organisation can take in changes, distribute them internally and return them to the higher-level context without losing its basic binding.

At the atomic level, this order appears as stable binding and a discrete spectral response. At the biological level, it becomes readable as integrated reaction, storage and sensory capacity. At the galactic level, it appears as long-term coupling of baryonic distribution, rotation, radiation, saturation and return flow. The forms differ, but the organisational rule remains the same:

intake
⇓
internal reorganisation
⇓
feedback
⇓
stabilised visibility

Saturation does not fundamentally interrupt this order. It concentrates the feedback until a point is reached at which release becomes necessary. Release radiation, ejection, pair formation, material re-coupling or the transition into a new visible organisational form then represent different expressions of the same coherent reorganisation. The Phraqtal Field therefore remains open to change on all scales while remaining sufficiently bound to carry history, form and material persistence.

1.6. Bound Overall Field States

A bound system organises itself from a central binding region, outer coupling nodes and a distributed outer response:

$$\mathcal{S} = \mathcal{B} + \sum_i w_i \mathcal{K}_i + \mathcal{A}. \tag{1}$$

$\mathcal{B}$ denotes the central binding region, $\mathcal{K}_i$ the outer coupling nodes with their effective contribution w_i, and $\mathcal{A}$ the connected outer response.

Organisational role	Overall atomic state	Overall galactic state
Central binding	Atomic nucleus	centrally condensed host region
Outer nodes	Bound electron modes	Satellite, gas and stellar nodes
Coupling regions	Shell and state families	differently bound outer regions
Outer response	Shell reaction and polarisability	Dynamic outer response

Tabelle 1: Scale-relative organisational roles of atom and galaxy.

1.7. Binding, Alignment and Repulsion in the Phraqtal Field

The Phraqtal Field does not distinguish between independent realities, but between different organisational responses of the same field. Binding, directed alignment, repulsion and radiation arise from the way in which field forms absorb, distribute and couple their internal tension to neighbouring organisations.

Gravitation refers to the large-scale binding response. It keeps distant field organisations within a common geometric order. Its effect is not limited to visible matter, but follows the entire effective binding, pre-tension and reorganisation structure of a system. In galaxies, this response becomes readable as the dynamic organisation extending beyond the baryonic distribution.

Magnetic phenomena, by contrast, form a locally directed alignment and twisting response. They occur when bound field organisations possess a coherent orientation of their internal circulation. Their external effect increases with the coherence C and the directed twisting component T:

$$\mathcal{M}_{\text{auen}} \propto C \cdot T.$$

The formula is an organisational relationship: without coherent alignment, a local twist remains largely unreadable externally. With increasing coherence, however, it can have an ordering effect across larger regions. Magnetic order is therefore not an additional substance in the field, but the directed readability of an internal field organisation.

Repulsion arises when two organisations do not possess a common coherent form of connection. Their respective feedback directions are then incompatible. The field does not form a common deepening of binding, but reorganises the boundary zone so that the incompatible orders are separated. Attraction and repulsion are therefore not opposing fundamental principles, but two results of the same question:

Is a common stable feedback possible?

If it is possible, binding and common organisation reinforce each other. If it is not possible, reorganisation leads to separation, alignment or release. This relationship applies from the microscopic domain to plasmas, stars, galaxies and galactic centres.

Within this order, radiation appears as the moving transmission of a field change. Matter stores bound field history; gravitation connects this history on a large scale; magnetic order aligns it locally and coherently; repulsion protects the inherent order of incompatible forms of binding. The Phraqtal Field brings these phenomena together as different readings of one continuous reorganisation.

1.8. Scale Cascade of the Phraqtal Field

The Phraqtal Field does not organise itself in isolated stages. Every bound structure arises from already existing field motion, develops its own internal coherence and can in turn become the carrier of larger organisational forms. This results in a scale cascade of bound and free field states:

$$
\begin{array}{c}
Q \\
\Downarrow \\
\Phi \\
\Downarrow \\
\text{free field motion} \\
\Downarrow \\
\text{minimal binding} \\
\Downarrow \\
\text{matter} \\
\Downarrow \\
\text{complex overall organisation}
\end{array}
$$

The individual scales differ in size, time order, binding energy and visibility. They nevertheless follow the same basic organisational sequence: field tension is bound, saturated, released, reorganised and coupled again.

1.8.1 The First Binding Stage

The first stable, materially coupling-capable organisation arises from minimally bound field modes. In the Phraqtal Field, the electron and positron form the first oppositely coupled pair organisation:

$$\Phi_{\text{frei}} \longrightarrow e^- e^+.$$

This stage connects free release tension with bound material capacity. It is not yet a complex matter organisation, but it already contains the prerequisites for binding, counter-feedback, visibility and further reorganisation.

From this first binding capacity, the first long-term stable forms of matter develop through quark organisations and proton–neutron bindings:

$$
\begin{array}{c}
e^- e^+ \\
\Downarrow \\
\text{quark organisation} \\
\Downarrow \\
\text{nucleons} \\
\Downarrow \\
\text{H, He}
\end{array}
$$

1.8.2 Hydrogen and Helium as the First Basic Matter Families

Hydrogen and helium form the first two dominant matter classes of the Phraqtal Field.

Hydrogen is the open initial form of matter:

$$H: \quad \text{minimal stable binding} + \text{high external coupling capacity}.$$

Helium is the first closed stable basic form:

$$He: \quad \text{integrated pair organisation} + \text{increased internal coherence}.$$

These two basic families continue the later logic of element organisation. Open states enable new couplings, bindings and reorganisations. Closed states enable long-term stabilisation and ordered overall integration.

1.8.3 From Atoms to Stellar and Galactic Organisations

With increasing complexity, atoms, molecules, planets, stars and galaxies do not become separate ontological categories. They form successive bound organisational stages:

$$\begin{array}{c} \text{atom} \\ \Downarrow \\ \text{molecule} \\ \Downarrow \\ \text{macroscopic matter} \\ \Downarrow \\ \text{planet} \\ \Downarrow \\ \text{star} \\ \Downarrow \\ \text{galaxy} \end{array}$$

A star is a highly condensed material reorganisation organisation. A galaxy integrates stars, gas, satellites, central saturation regions and a large-scale outer field into a higher-level form of binding.

The galaxy therefore takes on the same organisational roles that appear in the atom as the core region, outer binding, shell structure, reorganisation and neutralisation. Its specific form, however, takes shape within its own spatial and temporal regime order.

1.8.4 The Quasar as a Transition between Binding and New Formation

Within the scale cascade, the quasar occupies a special position. It is not the endpoint of galactic binding, but a highly saturated transition state. In the quasar region, central binding, maximum saturation, release radiation and matter formation come together directly:

$$
\begin{array}{c}
\text{galactic binding} \\
\Downarrow \\
\text{central saturation} \\
\Downarrow \\
\text{release} \\
\Downarrow \\
\text{new matter organisation}
\end{array}
$$

Through this, the quasar connects the internal history of a host galaxy with its future outer organisation. It can influence gas, star formation, satellite coupling and the later return of reorganised matter.

1.8.5 Observer Organisation as a Later Integration Stage

A biological observer is a highly complex bound field organisation that integrates matter, energy, memory and reorganisation capacity. Its observation is not an external addition to the Phraqtal Field, but another organisational stage within the same field:

$$
\begin{array}{c}
\text{field modulation} \\
\Downarrow \\
\text{sensory and measurement coupling} \\
\Downarrow \\
\text{bound processing} \\
\Downarrow \\
\text{integrated observation}
\end{array}
$$

The cascade therefore does not close as a circle in the geometric sense, but as a continuing feedback structure. Field motion forms matter; matter forms complex bound systems; complex systems read and reorganise field motion.

1.8.6 Shared Organisational Roles

The scale cascade is held together by six recurring roles:

$$
\left\{
\begin{array}{l}
\text{central binding,} \\
\text{outer coupling,} \\
\text{saturation,} \\
\text{release,} \\
\text{reorganisation,} \\
\text{visibility.}
\end{array}
\right.
$$

These roles appear in the atom, the galaxy, the quasar and the observer, each in its own form. The Phraqtal Field therefore does not describe a series of isolated objects, but the continuing organisation of a continuous Phraqtal Field in different bound and readable states.

1.9. Organisational State and Development Rules of the Phraqtal Field

The concepts developed above can be brought together into a common description of states. This description is not a complete field equation. It is the organisational coordinate language of the Phraqtal Field, used to describe atoms, galaxies, quasars and observers according to the same basic roles.

For every bound organisation X, we define the Phraqtal state

$$\mathfrak{P}_X = \left(K_X,\, O_X,\, \Sigma_X,\, C_X,\, R_X,\, q_{\mathrm{B},X},\, q_{\mathrm{A},X},\, \mathcal{V}_X\right).$$

The components have the following meaning:

$$\begin{aligned}
K_X &: \text{central binding strength,}\\
O_X &: \text{outer openness and coupling capacity,}\\
\Sigma_X &: \text{local saturation state,}\\
C_X &: \text{internal coherence,}\\
R_X &: \text{reorganisation readiness,}\\
q_{\mathrm{B},X} &: \text{spatial form of the outer organisation,}\\
q_{\mathrm{A},X} &: \text{dynamic activation state,}\\
\mathcal{V}_X &: \text{visibility in the respective observation regime.}
\end{aligned}$$

Different systems are therefore not compared by their absolute size, but by their position within the common organisational space:

$$\mathfrak{P}_{\mathrm{Atom}} \longleftrightarrow \mathfrak{P}_{\mathrm{Galaxy}} \longleftrightarrow \mathfrak{P}_{\mathrm{Quasar}}.$$

1.9.1 Binding and Openness

Central binding and outer openness form the first basic tension of every stable organisation:

$$\mathcal{B}_X = K_X - O_X.$$

A strongly bound state with too little outer openness cannot sufficiently transfer tension into new couplings. A state that is too open and has too little binding cannot maintain its coherence over time. Stability therefore does not arise from the maximum of a single quantity, but from their ordered balance:

$$0 < O_X < O_{\mathrm{krit},X}, \qquad K_X > K_{\mathrm{min},X}.$$

At the atomic level, this relationship describes the possibility of stable outer electron organisations; at the galactic level, it describes the ability to couple gas, satellites and outer field structures in an ordered way.

1.9.2 Saturation and Release

Saturation increases when central binding, inflow and internal compression increase more strongly than the available release capacity:

$$\frac{d\Sigma_X}{dt} \propto J_{\mathrm{in},X} + J_{\mathrm{verd},X} - J_{\mathrm{rel},X}.$$

Below a critical threshold, the tension remains integrated into the existing organisation:

$$\Sigma_X < \Sigma_{\mathrm{krit},X}.$$

When the threshold is reached or exceeded, reorganisation begins:

$$\Sigma_X \geq \Sigma_{\mathrm{krit},X} \quad \Longrightarrow \quad R_X \uparrow, \qquad q_{\mathrm{A},X} > 0.$$

Depending on the scale, release can appear as radiation, transition, particle formation, redistribution, outflow, matter formation or feedback.

1.9.3 Coherence and Visibility

A reorganisation becomes clearly visible only when it remains sufficiently coherent and is coupled to the observation regime:

$$\mathcal{V}_X = C_X \, \kappa_{X\to 0} \, \mathcal{R}_X \, \mathcal{M}_X.$$

Visibility therefore does not necessarily increase with the total field tension. A strong but incoherent organisation, or one coupled only weakly to the regime, can remain faintly visible. A smaller, locally coherent reorganisation, on the other hand, can produce a clear signal.

Within the Phraqtal Field, this explains why young stellar populations, quasars, radio sources or spectral lines become particularly visible: they combine release, coherence and suitable regime coupling.

1.9.4 Shape and Activation

The two Q-coordinates complete the organisational state. q_{B} describes the spatial form of the outer field response; q_{A} describes its dynamic activation level.

$$\left(q_{\mathrm{B}}, q_{\mathrm{A}}\right) = \left(\text{shape}, \text{switching}\right).$$

This means that two galaxies can have a similar total mass and nevertheless correspond to different element families. One may have a concentrated, activated outer organisation; the other a broadly integrated, non-activated overall organisation.

1.9.5 The Four Basic Developments

The common description of states gives rise to four fundamental directions of development:

$$\begin{aligned} \text{binding} &: O_X \longrightarrow K_X, \\ \text{openness} &: K_X \longrightarrow O_X, \\ \text{saturation and release} &: \Sigma_X \longrightarrow R_X, \\ \text{reintegration} &: R_X \longrightarrow C_X. \end{aligned}$$

Binding creates stable material organisation. Openness enables new couplings. Saturation forces release and reorganisation. Reintegration creates coherence again and a new stable overall form.

These four developments occur on all scales. They form the minimal dynamic grammar of the Phraqtal Field:

$$\begin{array}{c} \text{binding} \\ \Downarrow \\ \text{openness} \\ \Downarrow \\ \text{saturation} \\ \Downarrow \\ \text{release} \\ \Downarrow \\ \text{reintegration} \end{array}$$

The element–galaxy assignment is therefore not a list of external similarities. It is the assignment of common organisational states within this recurring sequence of development.

2. Visibility, Measurement and Regime Readability

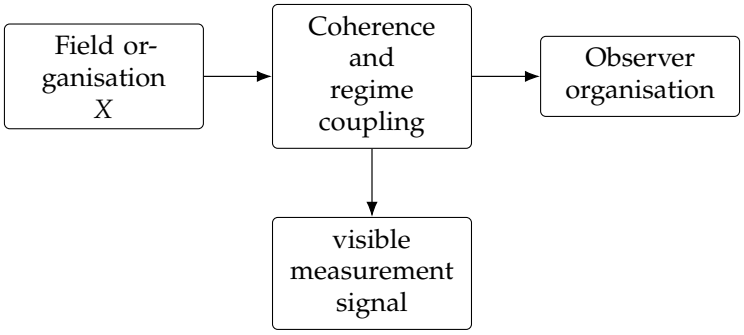


Abbildung 2: Visibility as a coupling event between a field organisation and a bound observer organisation.

Within the Phraqtal Field, an observer is not an entity standing outside the field. A measuring instrument, an eye, an atom, a detector and a living organism are themselves bound field organisations. Observation arises when incoming field motion couples with an already coherently bound organisation and produces a readable reorganisation there.

$$\begin{aligned}\text{incoming field modulation} \;+\; &\text{bound observer organisation}\\ &\longrightarrow \text{measurable reorganisation.}\end{aligned}$$

The observer therefore does not receive the complete reality of the Phraqtal Field. It couples to those modulation layers whose frequency, coherence, geometry, duration and tension strength are compatible with its own organisational form.

2.1.1 Visibility as a Coupling Event

Visibility is not an absolute property of an object. A field organisation becomes visible when it is sufficiently coherently connected to the coupling filter of an observation regime:

$$\mathcal{V}_{X\to O} = C_X\, \kappa_{X\to O}\, \mathcal{M}_X\, \mathcal{A}_O.$$

Here C_X denotes the coherence of the observed structure, $\kappa_{X\to O}$ its coupling to the observer, $\mathcal{M}_X$ the supporting modulation and $\mathcal{A}_O$ the observer organisation's capacity to receive and read out the signal.

A structure can therefore physically exist without being visible:

$$\mathcal{F}_X \neq 0, \qquad \mathcal{V}_{X\to O} \simeq 0.$$

This applies not only to distant or faint objects. Sub-regime-like field organisations, pre-coherent matter-formation zones and weakly coupled satellite structures can also be physically effective without appearing directly as independent objects in our regime.

2.1.2 Material Readability

Within the Phraqtal Field, matter is the form of field motion that has become sufficiently stably bound to couple repeatedly with other organisations. Its materiality does not consist in being separate from the field, but in the persistence of its binding:

$$\text{materiality} = \text{bound coherence} + \text{repeatable coupling capacity}.$$

An electron, an atom, a star or a galaxy is material because its internal organisation remains stable over a characteristic period and produces repeatable effects in other bound field organisations.

The visibility of a material structure, by contrast, is only a particular form of its readability. An

atom can become visible through spectral lines, a galaxy through starlight, gas or acceleration profiles; both nevertheless remain physically organised even where their direct optical visibility is low.

2.1.3 Resolution Depth

The physically describable depth of an observation is called resolution depth. It indicates how many organisational layers of a field motion an observer can distinguish separately:

$$\mathcal{D}_{\text{Auflsung}} = \mathcal{D}\left(\Delta x,\, \Delta t,\, \Delta \nu,\, C_O,\, \kappa_{X\to O}\right).$$

A low resolution depth makes a complex field organisation appear as a simple, point-like or smooth object. As resolution depth increases, internal modulations, transitions, coupling zones and substructures become readable.

An electron can appear in our regime as a minimal bound particle state. In the context of a finer sub-regime, the same organisation becomes readable as a more extended internal field structure. Correspondingly, a galaxy can appear in the super-regime as a sub-organisation of a larger supporting field motion.

2.1.4 Depth of Sensation

The depth of sensation is not identical to the technical resolution depth. It describes the qualitative range within which a bound living observer organisation distinguishes, integrates and experiences its coupled field modulations as a coherent whole.

It is not an additional physical substance and not an independent force. It arises at the level of a highly complex bound observer from the integration of numerous coupled modulation layers:

$$\text{depth of sensation} = \text{integrated readability}\left(\mathcal{D}_{\text{Auflsung}}, C_O, \mathcal{H}_O\right),$$

where $\mathcal{H}_O$ denotes the observer's internal organisational history.

The physically testable question remains which field modulations reach an observer and which reorganisations they trigger in their body or measuring device. Sensation itself is the internal, experienced side of this highly integrated observer organisation.

2.1.5 Consequence for the Phraqtal Field

The Phraqtal Field is neither a backdrop behind matter nor a mere image of reality. It is the physically real organisational basis in which matter is bound, radiation propagates freely, visibility is coupled and experience is integrated in complex observer organisations.

$$Q \longrightarrow \Phi \longrightarrow \begin{cases} \text{bound matter}, \\ \text{free radiation}, \\ \text{regime-dependent visibility}, \\ \text{integrated observation}. \end{cases}$$

The depth of what becomes visible or can be experienced therefore does not depend solely on the outer structure. It also depends on the coupling and resolution capacity of the bound organisation that reads this structure.

2.2. Visibility and Regime Coupling

Visibility arises through sufficient coherent coupling to the respective observation regime:

$$\text{visibility} = \text{coherence} \times \text{regime coupling}. \tag{2}$$

Field organisations may already exist before they become visible and technically evaluable as independent structures. On the galactic scale, this applies to gas accumulations, weakly coherent outer nodes and the development of young stellar populations. With increasing coherence and star formation, a structure appears in our regime as a visible dwarf galaxy.

The appearance and disappearance of outer nodes describe changes of state within a connected host organisation. An outer node can arise, couple strongly, be reorganised through tidal interaction and then be integrated back into the host galaxy.

2.3. Measurement as Coupled Field Reorganisation

Within the Phraqtal Field, an observer is never outside the event being observed. Every measuring device, every eye, every biological nervous system and every technical detector arrangement is itself a bound field organisation. Observation arises when an incoming tension front together with its internal modulation triggers a local reorganisation cascade in the detector.

incoming field modulation
⇓
local coupling
⇓
reorganisation of the detector
⇓
measurable signal

A measured signal is therefore not merely an image of an external property. It is the stable, repeatable response of a bound system to an incoming field change. Electron acceleration, ionisation, heating, pair formation, electrical voltage or an optical pulse describe different visible consequences of the same basic structure: the detector system absorbs the energy and orientation of the incoming modulation and reorganises itself through counteraction.

This counteraction is part of the measurement. A field organisation cannot absorb a change without slightly changing its own bound motion. The response spreads as a cascade into further coupled sub-organisations. In a technical sensor it becomes readable as current, voltage, a counting event or a spectral line. In a biological system it can be integrated as perception, orientation, a memory trace or sensation.

The visibility of an object is therefore not an absolute property. It depends on whether sufficient coupling exists between the object organisation and the respective detector organisation. For an observable response, at least four conditions must coincide:

$$\mathcal{V} = \mathcal{V}\left(E_{\text{in}}, C_{\text{obj-det}}, \Sigma_{\text{det}}, A_{\text{det}}\right).$$

Here E_{in} denotes the coupled field energy, $C_{\text{obj-det}}$ the coherence between object and detector, Σ_{det} the saturation state of the detector and A_{det} its resolution and response capacity. If one of these conditions is absent, an existing field organisation remains invisible along this observation path or can only be read indirectly.

The Phraqtal Field therefore distinguishes between the existence of an organisation and its visibility in a particular regime. Visibility is the successful translation of a field structure into a reorganisation of the observer. Material persistence, radiation, spectrum, measurement signal and sensation are not separate realities, but different depths of the same coupled readability.

2.4. Release Radiation, Spectra and Regime Visibility

Release fields occur in every bound organisation within the Phraqtal Field. They are the continuing response of a system to internal tension, external coupling and structural reorganisation. Their observable radiation depends not only on the existing field tension, but also on the strength, coherence and regime coupling of the respective release.

For a bound system X, the measurable release radiation can be written as a working quantity:

$$I_{\text{obs},X} = \mathcal{R}_X \, C_X \, \kappa_{X\to 0} \, \mathcal{P}_X.$$

Here $\mathcal{R}_X$ denotes the release rate, C_X the coherence of the reorganisation, $\kappa_{X\to 0}$ the coupling to our regime R_0 and $\mathcal{P}_X$ the spectral projection into an observable energy range.

The equation separates four states that would otherwise be easily mixed together in an observation:

$\mathcal{R}_X > 0,\ C_X$ small	:	release present, but weakly visible,
$\mathcal{R}_X > 0,\ C_X$ high, $\kappa_{X\to 0}$ small	:	coherent organisation outside visibility,
$\mathcal{R}_X > 0,\ C_X$ high, $\kappa_{X\to 0}$ high	:	strongly visible release,
$\mathcal{R}_X$ changes	:	observable change of state.

2.4.1 Atomic Spectra as an Indicator of State

In the atom, release radiation appears as a spectral response to the reorganisation of bound electron modes. The emission marks the transition between two stable field organisations:

$$\mathcal{E}_n \longrightarrow \mathcal{E}_m + \gamma_{nm}.$$

Within the Phraqtal Field, the observed photon is the regime-coupled representation of the reorganisation. It is not separate from the bound state, but carries its energy difference, direction and coherence outward.

A closed organisation such as helium has high internal coherence and limited external reorganisation capacity. Its spectral response therefore arises under different conditions than that of an open or partly open element. Hydrogen, beryllium, magnesium or calcium each possess their own coupling and reorganisation spaces, expressed in the variety of their possible state transitions.

2.4.2 Galactic Spectra as an Indicator of State

On the galactic scale, the same basic organisation appears in a variety of observable signals: star formation, ionised gas, radio emission, X-ray radiation, infrared radiation, jets, outflows and the optical visibility of young stellar populations. These signals do not arise independently of one another. They are different projections of the release and reorganisation dynamics of a galaxy.

$$(q_{\mathrm{B}}, q_{\mathrm{A}}) = (\text{shape}, \text{switching}) .$$

This means that two galaxies can have a similar total mass and nevertheless correspond to different element families. One may have a concentrated, activated outer organisation; the other may have a broadly integrated, non-activated overall organisation.

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The common description of states gives rise to four fundamental directions of development:

$$\begin{aligned} \text{binding} &: O_X \longrightarrow K_X, \\ \text{openness} &: K_X \longrightarrow O_X, \\ \text{saturation and release} &: \Sigma_X \longrightarrow R_X, \\ \text{reintegration} &: R_X \longrightarrow C_X. \end{aligned}$$

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These four developments occur on all scales. They form the minimal dynamic grammar of the Phraqtal Field:

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The element–galaxy assignment is therefore not a list of external similarities. It is the

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3. Visibility, Measurement and Regime Readability

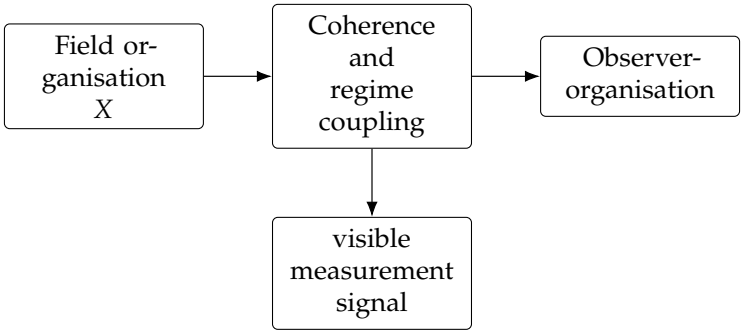


Abbildung 3: Visibility as a coupling event between a field organisation and a bound observer organisation.

3.1. Observer, Visibility and the Depth of Field Readability

Within the Phraqtal Field, an observer is never an entity standing outside the observed event. Every measuring device, every eye, every biological nervous system and every technical detector arrangement is itself a bound field organisation. Observation arises when an incoming field motion couples to an already coherently bound organisation and produces a readable reorganisation there.

$$\begin{aligned} &\text{incoming field modulation} \;+\; \text{bound observer organisation} \\ &\qquad\qquad\qquad\qquad \longrightarrow\; \text{measurable reorganisation.} \end{aligned}$$

The observer therefore does not receive the complete reality of the Phraqtal Field. It couples to those modulation layers whose frequency, coherence, geometry, duration and tension strength are compatible with its own organisational form.

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Visibility is not an absolute property of an object. A field organisation becomes visible when it is sufficiently coherently connected to the coupling filter of an observation regime:

$$\mathcal{V}_{X\to O} = C_X\, \kappa_{X\to O}\, \mathcal{M}_X\, \mathcal{A}_O .$$

Here C_X denotes the coherence of the observed structure, $\kappa_{X\to O}$ its coupling to the observer, $\mathcal{M}_X$ the supporting modulation and $\mathcal{A}_O$ the observer organisation's capacity to receive and read out the signal.

A structure can therefore physically exist without being visible:

$$\mathcal{F}_X \neq 0, \qquad \mathcal{V}_{X\to O} \simeq 0.$$

This applies not only to distant or faint objects. Sub-regime-like field organisations, pre-coherent matter-formation zones and weakly coupled satellite structures can also be physically effective without appearing directly as independent objects in our regime.

3.1.2 Material Readability

Within the Phraqtal Field, matter is the form of field motion that has become sufficiently stably bound to couple repeatedly with other organisations. Its materiality does not consist in being separate from the field, but in the persistence of its binding:

$$\text{materiality} = \text{bound coherence} + \text{repeatable coupling capacity}.$$

An electron, an atom, a star or a galaxy is material because its internal organisation remains stable over a characteristic period and produces repeatable effects in other bound field organisations.

The visibility of a material structure, by contrast, is only a particular form of its readability. An atom can become visible through spectral lines, a galaxy through starlight, gas or acceleration profiles; both nevertheless remain physically organised even where their direct optical visibility is low.

3.1.3 Resolution Depth

The physically describable depth of an observation is called resolution depth. It indicates how many organisational layers of a field motion an observer can distinguish separately:

$$\mathcal{D}_{\text{Auflsung}} = \mathcal{D}\left(\Delta x,\, \Delta t,\, \Delta \nu,\, C_O,\, \kappa_{X\to O}\right).$$

A low resolution depth makes a complex field organisation appear as a simple, point-like or smooth object. As resolution depth increases, internal modulations, transitions, coupling zones and substructures become readable.

An electron can appear in our regime as a minimal bound particle state. In the context of a finer sub-regime, the same organisation becomes readable as a more extended internal field structure. Accordingly, a galaxy can appear in the super-regime as a sub-organisation of a larger supporting field motion.

This difference is written as the closure index

$$\Gamma_E = \frac{N_{\text{occupied}}}{N_{\max}} \tag{3}$$

For the first atomic shell, the following applies:

$$\Gamma_{\mathrm{H}} = \frac{1}{2}, \qquad \Gamma_{\mathrm{He}} = 1. \tag{4}$$

$\Gamma_E = 1$ denotes a completely closed coupling architecture. The closure index does not describe the absolute size of a system, but the degree to which its outer binding structure has reached its stable occupancy state.

In the galactic regime, the corresponding quantity is defined as the stabilisation index

$$\Gamma_G = \frac{N_{\mathrm{integrated},G}}{N_{\mathrm{eff},G}} \tag{5}$$

Here, $N_{\mathrm{integrated},G}$ denotes the share of outer substructures that is coherently integrated into the overall organisation of the host galaxy over the long term. $N_{\mathrm{eff},G}$ denotes the entire effective outer occupancy consisting of satellites, gas structures, star-formation zones and other coupling-capable outer modes.

A high value of Γ_G describes a galaxy with a stabilised relationship between its centre and outer organisation. A lower value describes a galaxy whose outer occupancy is still open, newly forming, only weakly coupled or highly reorganisable.

Tabelle 2: Early atomic references and their galactic organisational roles.

Reference	Atomic state	galactic organisational role	Closure
Hydrogen	open single binding	open outer coupling, ongoing new formation and reorganisation	$\Gamma < 1$
Helium	closed pair binding	stabilised inner–outer coupling, high integration coherence	$\Gamma \approx 1$

The helium class should not be equated with a completely quiescent galactic system. Release continues to take place. The difference is that the release is integrated into an already strongly stabilised overall architecture. A helium-like galaxy class can therefore carry both peak and diffuse profiles. What matters is not the shape q_{B} alone, but the combination of shape, activation and stabilisation:

$$\mathcal{H}_G = \left(q_{\mathrm{B}}, q_{\mathrm{A}}, T, \Gamma_G, \sigma_{\mathrm{ext}}\right). \tag{6}$$

This gives rise to four basic early Phraqtal Field states:

The last state is particularly important for the Phraqtal Field: A galaxy can be both highly integrated and dynamically active. Stability does not mean stillness. It means that ongoing release, visibility, satellite coupling and reorganisation take place within a sustainable bound overall structure.

The extended state coordinate is therefore

$$\mathcal{S}_{E,G} = \left(\mathcal{K}_{\mathrm{centre}}, \mathcal{A}, \Gamma, W, S, Q, \chi, \Pi, q_{\mathrm{B}}, q_{\mathrm{A}}, T, \sigma_{\mathrm{ext}}, N_{\mathrm{eff}}\right). \tag{7}$$

Tabelle 3: Early Phraqtal Field states arising from openness, closure and dynamic activation.

State	Atomic reference	Galactic reading
open, non-activated	H-like	predominantly open outer coupling without a developed transition scale
open, activated	H-like	open outer organisation with dynamic release switching
closed, non-activated	He-like	stabilised overall structure without an active transition scale
closed, activated	He-like	stabilised overall structure with coherent dynamic transition organisation

Helium therefore adds the previously missing axis of the first complete closure to the element–galaxy assignment. Hydrogen represents the early open binding; helium represents its first complete stabilisation. All later elements and galaxy classes can be placed between, above or alongside these two basic forms.

3.2. From Hydrogen and Helium to Material Diversity

The material development of the Phraqtal Field begins with minimal coupling-capable field forms. In release tension fields of high saturation, electron–positron pairs arise as the first stable, externally effective counter-organisations. They mark the transition from a field state carried mainly by radiation and tension to a materially connectable order of binding.

From this first coupling-capable stage, the further condensation of matter organisation develops:

$$
\begin{array}{c}
\text{release field} \\
\Downarrow \\
e^- e^+ \\
\Downarrow \\
\text{quark organisation} \\
\Downarrow \\
\text{protons and neutrons} \\
\Downarrow \\
\text{hydrogen and helium}
\end{array}
$$

Hydrogen forms the first open, widely coupling basic material order. It is mobile, capable of binding and highly able to connect to larger material organisations. Helium, by contrast, represents the first especially coherent, closed and energetically stable elemental order. Within the Phraqtal Field, its high abundance follows from its role as an early stable condensation form after the open hydrogen organisation.

$$\text{H} \ : \ \text{open material coupling}, \qquad \text{He} \ : \ \text{closed coherent stabilisation}.$$

With the formation of stars, regions arise in which these elementary bindings are condensed, heated and reorganised again. Stars are not merely accumulations of already finished

matter. They form long-lasting transformation centres in which the open binding capacity of hydrogen and the stability of helium are transferred into new equilibria. In this way, the diversity of element organisations increases.

Heavy elements appear as later, more highly structured forms of material storage. They possess more binding centres, more possible coupling states and a greater internal organisational density. In this way, they expand the ability of the Phraqtal Field to form stable structures, chemical differentiation and complex feedback spaces.

The development from light to heavy matter does not follow a simple linear accumulation. Each new elemental order arises from the interaction of binding, saturation, release and renewed coupling:

$$\begin{array}{c} \text{binding} \\ \Downarrow \\ \text{condensation} \\ \Downarrow \\ \text{saturation} \\ \Downarrow \\ \text{release} \\ \Downarrow \\ \text{new material order} \end{array}$$

The same organisational pattern subsequently appears on larger scales. Atoms become molecules, molecules become materials, materials become planets and stars, and star systems become galaxies. The material diversity of the universe is therefore the progressive storage and differentiation of field history in ever-new, yet mutually coupled, forms of binding.

3.3. Periodic Organisational Families in the Phraqtal Field

The chemical elements do not form an unordered sequence. Their properties repeat because open and closed electron shells alternate in a periodic organisational sequence. Within the Phraqtal Field, this periodicity is read as the recurrence of bound organisational roles across different scales.

The atomic shell sequence can be described by a state family

$$\mathcal{P}_E = (n,\ \Gamma_E,\ \mathcal{A}_E,\ \chi_E,\ Q_E) \tag{8}$$

where n denotes the shell order, Γ_E the degree of closure, $\mathcal{A}_E$ the occupancy state, χ_E the response capacity and Q_E the transition relation between inner and outer organisation.

The basic atomic sequence is:

$$\begin{aligned} \text{open binding} &\longrightarrow \text{first closure} \\ &\longrightarrow \text{multi-shell openness} \longrightarrow \text{multi-shell closure}. \end{aligned} \tag{9}$$

Hydrogen and helium form the first complete organisational period:

$$\mathrm{H}(1s^1) \longrightarrow \mathrm{He}(1s^2). \tag{10}$$

Hydrogen stands for the first open bound coupling. Helium stands for the first complete closure. With beryllium, a new outer occupancy begins above the closed helium structure:

$$\mathrm{He}(1s^2) \longrightarrow \mathrm{Be}(1s^2\,2s^2). \tag{11}$$

Neon closes this second period as a stable outer shell organisation. Magnesium opens another outer occupancy, while argon closes this period.

Tabelle 4: Atomic periods as recurring Phraqtal Field organisational roles.

Role	Examples	atomic structure	Phraqtal Field function
first openness	H	$1s^1$	open initial coupling
first closure	He	$1s^2$	stable pair binding
renewed openness	Be, Mg, Ca	outer s occupancy	new outer coupling architecture
multi-shell closure	Ne, Ar	closed outer shell	integrated stable outer organisation

In the galactic regime, the same sequence does not appear as an electron shell, but as a sequence of central binding, outer occupancy, transition activation and integration. The galactic period coordinate is

$$\mathcal{P}_G = \left(\mathcal{K}_{\mathrm{centre},G},\, \Gamma_G,\, q_\mathrm{B},\, q_\mathrm{A},\, T,\, N_{\mathrm{eff},G}\right). \tag{12}$$

The shell order n does not correspond to a simple number of galaxies, satellites or stars. It corresponds to the stage at which a new outer coupling architecture is formed. A galactic period begins with an open outer formation and satellite organisation. It stabilises through coherence, visibility and integration and reaches a closed overall state. After that, a new occupancy architecture can develop farther out.

$$\begin{aligned} &\text{open outer coupling} \\ &\quad\longrightarrow \text{stabilised overall binding} \\ &\quad\longrightarrow \text{new outer occupancy} \\ &\quad\longrightarrow \text{multi-stage integrated outer organisation.} \end{aligned} \tag{13}$$

Bimodality determines how this outer organisation appears in space. The peak regime describes a locally concentrated outer response. The diffuse regime describes a widely distributed outer response. The Q-switch, by contrast, determines whether the transition organisation between the centre and the outer region has already developed dynamically.

A galactic period can therefore be determined by three independent state questions:

$$\left(\Gamma_G,\, q_\mathrm{B},\, q_\mathrm{A}\right). \tag{14}$$

The element–galaxy assignment thereby becomes periodic without reducing a galaxy to a single atom. Hydrogen-like and helium-like classes mark the first open and the first

Tabelle 5: Galactic organisational families based on degree of closure, bimodality and activation.

Galactic family	State characteristics	Organisational role
open peak family	$\Gamma_G < 1$, peak, variable activation	locally concentrated new formation and outer coupling
closed peak family	$\Gamma_G \approx 1$, peak, activated or stable	compact and coherent overall binding
open diffuse family	$\Gamma_G < 1$, diffuse, variable activation	widely distributed outer occupancy and ongoing reorganisation
closed diffuse family	$\Gamma_G \approx 1$, diffuse, activated or stable	large-scale integrated release organisation

closed overall organisation. Neon-like and argon-like classes mark later stable outer closures. Beryllium-like, magnesium-like and calcium-like classes each mark new outer occupancy phases.

The Phraqtal Field thus acquires a periodic order of bound systems:

$$\mathcal{P}_E \longleftrightarrow \mathcal{P}_G. \tag{15}$$

The left-hand side describes the periodic sequence of atomic shell organisation. The right-hand side describes the recurring sequence of galactic central binding, outer coupling, release, activation and integration.

3.4. Abundance Order and Dominant Galactic Families

In addition to the structural and state axes, the element–galaxy assignment also has an abundance axis. Hydrogen and helium arise as the first bound elements and form the dominant early forms of matter. Their organisational roles therefore do not have to appear exclusively in rare specialised objects within the Phraqtal Field, but can recur in the most common stable galactic overall states.

For an element E, its material abundance role is denoted by

$$\mathcal{H}_E = \frac{N_E}{\sum_j N_j} \tag{16}$$

Here, N_E does not describe only the counted amount of an element, but its role in the overall organisation of bound matter. Hydrogen and helium provide the initial structure from which later star, gas, planet and element formation develops.

The galactic equivalent is the abundance of an organisational family:

$$\mathcal{H}_G(\mathcal{F}) = \frac{N_G(\mathcal{F})}{N_{G,\text{total}}}, \tag{17}$$

where $\mathcal{F}$ denotes a Phraqtal Field family determined by profile shape, activation and degree of stabilisation.

The bimodal analysis shows that galactic release organisation is not distributed evenly across all possible forms. Instead, preferred families occur:

$$\begin{aligned} \mathcal{H}_G(\text{Peak}) &\approx 0.62, \\ \mathcal{H}_G(\text{Transition}) &\approx 0.12, \\ \mathcal{H}_G(\text{Diffuse}) &\approx 0.26. \end{aligned} \tag{18}$$

The peak family is therefore the most common of the galactic field forms determined so far. It describes a locally concentrated outer release organisation. The diffuse family forms a second, less frequent but clearly distinguishable large-scale organisational form. The transition family connects both states.

Within the Phraqtal Field, the abundance order is read as an expression of preferred bound stability states:

$$\text{common atomic basic forms} \longleftrightarrow \text{common galactic basic families.} \tag{19}$$

Hydrogen represents the most common open initial coupling. Helium represents the most common early closed stabilisation. The galactic assignment follows the same basic distinction:

Tabelle 6: Abundance roles of early forms of matter and galactic organisational families.

Form of matter	atomic organisational role	galactic abundance role
Hydrogen	dominant open basic binding	common open or newly reorganising outer coupling
Helium	dominant early closure	common stabilised inner–outer overall binding
heavier elements	later differentiated forms of binding	specialised multi-shell galaxy families

The assignment of hydrogen and helium therefore initially takes place at the level of dominant organisational roles, not through one individually named galaxy. Hydrogen-like galactic families mainly carry open, broadly reorganisable outer couplings. Helium-like families carry more closed and integrated overall structures.

The activation axis refines this abundance order. A common galactic family can occur in a non-activated, transition or activated state:

$$\mathcal{H}_G(\mathcal{F}) = \mathcal{H}_G\left(\Gamma_G,\, q_B,\, q_A,\, T\right). \tag{20}$$

This makes it understandable why a common galactic form does not always have the same visible dynamics. The basic organisational family can remain intact while its release state, visibility, satellite coupling and transition scale change.

The periodicity of the elements and the preferred abundances of galactic regimes are therefore two sides of the same Phraqtal Field order: bound systems are not distributed randomly, but favour recurring states of openness, closure, outer occupancy and dynamic reorganisation.

4. Atomic and Galactic Organisational Roles

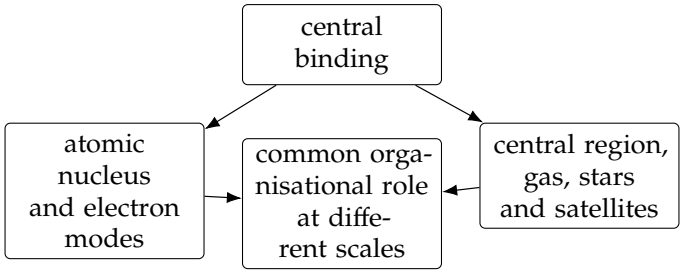


Abbildung 4: The atom–galaxy assignment compares organisational roles, not absolute sizes or orbits.

4.1. Element–Galaxy Assignment

The element–galaxy assignment classifies organisational signatures. Hydrogen-like and helium-like galaxies form the first stable basic classes; more complex elements correspond to further differentiated forms of binding, outer organisation and reorganisation. Visible dwarf galaxies are clearly coupled outer nodes of a more extensive galactic outer organisation.

For quantitative comparative work, the outer atomic structure and the galactic additional response are normalised to their own characteristic radii. The galactic additional component is:

$$g_{\mathrm{add}}(r) = g_{\mathrm{obs}}(r) - g_{\mathrm{bar}}(r). \tag{21}$$

Its positive cumulative outer organisation is written as

$$C_{\mathrm{Gal}}(r) = \frac{\int_0^r r' \, [g_{\mathrm{add}}(r')]_+ \, dr'}{\int_0^{R_{\mathrm{out}}} r' \, [g_{\mathrm{add}}(r')]_+ \, dr'}. \tag{22}$$

The first comparative signature consists of

$$W = \log_{10}\!\left(\frac{R_{90}}{R_{10}}\right), \qquad Q = \frac{R_{90}}{R_{\mathrm{eq}}}, \qquad \sigma_{\mathrm{ext}} = \left.\frac{g_{\mathrm{add}}}{g_{\mathrm{obs}}}\right|_{R_{90}}. \tag{23}$$

W describes the width of the outer organisation, Q its range beyond the coupling transition and σ_{ext} the strength of the outer dynamic response. Ionisation energy, polarisability and temporal and precessional relationships complete this spatial assignment.

4.2. Length and Time Relations of Element–Galaxy Candidates

The element–galaxy assignment is formulated through dimensionless organisational ratios. The absolute length scales of atomic shells and galactic visibility radii differ, as expected, by many orders of magnitude. What matters, therefore, is the scaling relation between an

atomic reference radius $r_{90,E}$ and the galactic reference radius $R_{90,G}$:

$$\mathcal{L}_{E,G} = \frac{R_{90,G}}{r_{90,E}}. \tag{24}$$

Here $r_{90,E}$ denotes the radius containing 90 % of the outer electron density under consideration. Correspondingly, $R_{90,G}$ denotes the radius within which 90 % of the galactic visibility or structure profile used is located. The relation therefore describes not merely a comparison of diameters, but the extent of the respective coupling-capable outer organisation.

Tabelle 7: Preliminary length relations of selected element–galaxy candidates.

Element mode	Galaxy	$r_{90,E}$ [Å]	$R_{90,G}$ [kpc]	$\mathcal{L}_{E,G}$
H, $1s$	UGC 01230	1.41	32.98	7.2×10^{30}
Be, $2s^2$	NGC 5033	2.23	40.73	5.6×10^{30}
Ne, outer shell	UGC 01230	0.86	32.98	1.2×10^{31}
Mg, $3s^2$	NGC 5033	2.67	40.73	4.7×10^{30}
Ar, outer shell	NGC 5033	1.33	40.73	9.4×10^{30}
Ca, $4s^2$	NGC 2955	3.36	32.54	3.0×10^{30}

The candidates lie within a common length corridor of approximately $10^{30.5}$ to 10^{31}. Within the Phraqtal Field, this corridor denotes the transfer of an outer bound coupling organisation from the atomic to the galactic regime. The spread also contains the differences between core binding, shell structure, visibility profile and release state.

For the time relation, a distinction must be made between the propagation time of a field reorganisation and the internal organisation time of a system. The front-bound time is

$$\tau_c = \frac{R}{c}. \tag{25}$$

If the maximum possible reorganisation speed in all regimes is determined by c, then for corresponding atomic and galactic reference radii:

$$\frac{\tau_{c,G}}{\tau_{c,E}} = \frac{R_{90,G}}{r_{90,E}} = \mathcal{L}_{E,G}. \tag{26}$$

The internal organisation time, by contrast, depends on the respective dynamics. For galactic motion structures, it can approximately be written as

$$\tau_{\mathrm{org},G} \sim \frac{R}{v} \tag{27}$$

For a complete orbit, this gives

$$T_{\mathrm{orb},G} = \frac{2\pi R}{v}. \tag{28}$$

In the atomic regime, the corresponding time does not describe a classical electron orbit. It denotes the time scale of a bound phase, density and coupling organisation. This gives the

hierarchy

$$\tau_c < \tau_{\text{org}} < \tau_{\text{macro}}, \tag{29}$$

where τ_{macro} denotes the time of large-scale reorganisation, release, satellite integration or structural new formation.

Within this framework, precession is understood as a slow modulation of a bound basic organisation. A dimensionless precession quantity can be defined by

$$\Pi = \frac{\Omega_{\text{prec}}}{\Omega_{\text{org}}} \tag{30}$$

Within the Phraqtal Field, Π relates the slow realignment of a system to its primary binding or coupling motion. In galaxies, this concerns, for example, orbital precession, nodal precession, bar, warp or satellite structures. In the atomic regime, it concerns the phase organisation of bound states. Precession therefore does not connect identical motion patterns, but the same hierarchical role: a slow reorganisation within a stable basic binding.

4.3. Coupling Capacity and Outer Occupancy

In addition to length, time and precession, the element–galaxy assignment also includes the number and stability of outer bound structures. In the atomic regime, this organisation appears as electron occupancy. In the galactic regime, it appears as a hierarchical distribution of satellite galaxies, gas accumulations, stellar populations, outer disc structures and other coupling-capable sub-organisations.

The core of an atom does not merely determine the attraction of individual electrons. It establishes the entire binding architecture: the possible number of stable electron modes, their shell structure, their energetic order and their transition possibilities. In the same way, the central bound region of a galaxy establishes the large-scale coupling architecture. This includes the inner baryonic centre of mass, the surrounding gas, the star-formation zones, the satellite population and the release structures located farther out.

For an atom, electron occupancy can be written as

$$N_e = \sum_j n_j \tag{31}$$

where n_j denotes the occupancy of the respective bound state j. Within the Phraqtal Field, the galactic equivalent is not captured solely as the number of visible dwarf galaxies. What matters is the effective number of coupling-capable outer organisations:

$$N_{\text{eff},G} = \sum_i C_i. \tag{32}$$

Here C_i denotes the coupling strength of structure i to the host galaxy. In particular, it includes its binding, coherence, spatial position, relative motion, gas density and visibility in the observed regime. A structure with high coupling contributes permanently to the galactic overall organisation; a weakly coupled structure initially remains peripheral, temporarily invisible or is integrated during a later reorganisation phase.

The observable number of outer structures is therefore only one part of the total occupancy:

$$N_{\mathrm{obs},G} = \sum_i \Theta(C_i - C_{\mathrm{krit}}) \,, \tag{33}$$

where Θ is the Heaviside function and C_{krit} is the coupling threshold of the observation regime. In general,

$$N_{\mathrm{obs},G} \leq N_{\mathrm{eff},G}. \tag{34}$$

A young dwarf galaxy or gas accumulation can therefore already be a real sub-organisation of the host galaxy, even though it does not yet produce a clear visible signal in our regime. Its structure is present, but its coherence and radiative coupling capacity are still below the visibility threshold. As the gas becomes denser and a young stellar population forms, regime coupling increases. The previously only weakly visible or invisible structure appears as an observable satellite, star-formation region or outer node.

This development sequence can be represented as

$$\text{formation zone} \longrightarrow \text{bound pre-structure} \longrightarrow \text{coupling-capable visible structure} \longrightarrow \text{integration} \tag{35}$$

Integration takes place through long-term interaction with the host field. Tidal effects, loss of orbital energy, gas transfer and structural mixing cause an outer organisation to gradually merge into the overall structure of the host galaxy.

In the atomic regime, this corresponds to a reorganisation of bound states: electron modes are absorbed, released or transferred into other bound configurations. In the galactic regime, it corresponds to the formation, becoming visible, binding, rearrangement and integration of outer substructures. The assignment therefore concerns the common organisational rule according to which a central bound state carries a variable, but not arbitrary, outer occupancy.

The chemical identity of an element and the structural class of a galaxy are both determined in this framework by the connection between central binding and outer occupancy. A larger or more complex outer occupancy does not necessarily mean a larger visible galaxy. It means a more developed coupling architecture with more possible stable, metastable, or integrable partial states.

4.4. Outer Field Response and Release Organisation

A bound state is not determined only by its internal structure. It also has an outer response capability. In the atomic regime, this appears in polarizability and in the possibility of reorganising the electron shell under external excitation. In the galactic regime, it appears in the additional acceleration organisation outside the baryon-dominated region.

For the atom, the response capability is described by a polarizability α_E. To connect it with the extent of the outer shell, the dimensionless response coordinate

$$\chi_E = \frac{\alpha_E}{r_{90,E}^3} \tag{36}$$

is used. It describes how strongly the outer electron organisation can be reorganised relative

to its own coupling volume.

In the galactic regime, the outer field response is described by the part of the observed acceleration that is not captured by the local baryonic component at the outer reference radius $R_{90,G}$:

$$\sigma_{\mathrm{ext},G} = \left.\frac{g_{\mathrm{obs}} - g_{\mathrm{bar}}}{g_{\mathrm{obs}}}\right|_{R_{90,G}}. \tag{37}$$

Here, g_{obs} is the acceleration determined from rotational motion, and g_{bar} is the baryonic acceleration reconstructed from gas, disk, and bulge. In the Phraqtal Field, $\sigma_{\mathrm{ext},G}$ describes the share of the outer release organisation that is effective for the observed galactic dynamics.

The atomic quantity χ_E and the galactic quantity $\sigma_{\mathrm{ext},G}$ have different units and different direct measurement paths. Their common role lies in the response of a bound overall system to its outer field organisation. Both quantities capture how far the observable dynamics extend beyond an exclusively local description based on the nucleus or baryons.

Tabelle 8: Preliminary response coordinates of selected elements and galactic comparison quantities.

Element	χ_E	Galactic reference	$\sigma_{\mathrm{ext},G}$
H	0.239	SPARC total sample, median	0.780
Be	0.502	NGC 5033	–
Ne	0.631	UGC 01230	–
Mg	0.556	NGC 5033	–
Ar	0.696	NGC 5033	–
Ca	0.611	NGC 2955	–

For the SPARC sample, the median of the outer acceleration fraction is

$$\widetilde{\sigma}_{\mathrm{ext}} = 0.780. \tag{38}$$

Thus, at the outer reference radius, a large share of the observed dynamics is, on average, carried by the nonlocal field organisation. The strength of this share is not identical to the geometric extent of a system. Galaxies with similar profile widths can have different outer release fractions, while galaxies with similar outer field responses can carry different visibility and satellite structures.

In the Phraqtal Field, release takes place continuously in all galaxies. Its measurability depends on its strength, coherence, geometry, and coupling to the observing regime. A compact and strongly concentrated release produces a clearer visible or high-energy signature. A diffuse release, on the other hand, remains visible mainly as a distributed dynamic response.

The overall galactic response can therefore be written in the working form

$$g_{\mathrm{obs}}(r) = g_{\mathrm{bar}}(r) + \Delta g_{\mathrm{ent}}(r) \tag{39}$$

The term $\Delta g_{\mathrm{ent}}(r)$ denotes the additional acceleration arising from the nonlocal organisation of the bound galactic field state. It represents the observed dark-matter phenomenology as a

release and coupling organisation.

The element–galaxy mapping therefore gains another working axis:

$$\mathcal{S}_{E,G} = (W,\ S,\ Q,\ \chi,\ \Pi,\ \sigma_{\mathrm{ext}})\,. \tag{40}$$

Here, W denotes the normalised profile width, S the concentration, Q the transition relation between inner and outer organisation, χ the response coordinate, Π the precession relation, and σ_{ext} the outer release fraction. This state coordinate makes it possible to classify elements and galaxies not by a single feature, but by their complete bound organisational signature.

4.5. Transition Zones between Inner Binding and Outer Organisation

Bound systems do not have a hard outer boundary. Between the strongly bound inner region and the outer coupling organisation lies a transition zone. In this zone, inner binding is not removed, but the ability to reorganise externally becomes increasingly decisive.

In the atomic regime, the transition zone lies between the electron density near the nucleus and the outer valence organisation. For multi-shell atoms, a transition radius $r_{\mathrm{eq},E}$ can be defined at which the contributions of the inner and outer electron organisations are equal:

$$\rho_{\mathrm{inner}}(r_{\mathrm{eq},E}) = \rho_{\mathrm{outer}}(r_{\mathrm{eq},E}). \tag{41}$$

The corresponding dimensionless atomic transition quotient is

$$Q_E = \frac{r_{90,E}}{r_{\mathrm{eq},E}}. \tag{42}$$

A large value of Q_E means that the outer coupling-capable shell extends far beyond the inner binding zone. A smaller value describes a more compact connection between the nucleus, inner shells, and outer shell.

In the galactic regime, the corresponding transition zone lies at the radius $R_{\mathrm{eq},G}$ where the baryonic acceleration organisation and the outer release organisation become equally strong:

$$g_{\mathrm{bar}}(R_{\mathrm{eq},G}) = \Delta g_{\mathrm{ent}}(R_{\mathrm{eq},G}). \tag{43}$$

With

$$\Delta g_{\mathrm{ent}}(r) = g_{\mathrm{obs}}(r) - g_{\mathrm{bar}}(r) \tag{44}$$

the galactic transition quotient follows as

$$Q_G = \frac{R_{90,G}}{R_{\mathrm{eq},G}}. \tag{45}$$

Q_G therefore describes how far the visible and dynamically coupled galactic outer organisation extends beyond the transition point. In the Phraqtal Field, this region is not empty space, but the region in which gas, stars, satellites, radiation, and the release field jointly carry the host galaxy's outer overall organisation.

Tabelle 9: Preliminary transition relations of selected atomic and galactic candidates.

Regime	System	Transition quotient	Value
Atomic	Ne, inner/outer shell	Q_E	5.79
Atomic	Ar, inner/outer shell	Q_E	3.29
Galactic	NGC 5033	Q_G	4.02
Galactic	SPARC sample, median	Q_G	6.40

The preliminary proximity of the Ar relation

$$Q_{\mathrm{Ar}} = 3.29 \tag{46}$$

and the relation determined for NGC 5033

$$Q_{\mathrm{NGC\,5033}} = 4.02 \tag{47}$$

classifies NGC 5033 as an argon-like transition class among the candidates considered so far. This classification adds to the previously identified similarity of the normalised outer profiles. It is therefore not based on a single size comparison, but on the shared position of the transition between inner binding and outer coupling organisation.

The transition is dynamic. In a young or strongly reorganised galaxy, $R_{\mathrm{eq},G}$ can shift because the gas distribution, star formation, satellite coupling, and release strength change. The atomic transition organisation likewise changes during ionisation, recombination, excitation, or chemical bonding. The transition scale is therefore a state quantity of the overall system.

For the further classification, the quotient Q is used together with the other organisational coordinates:

$$\mathcal{S}_{E,G} = (W,\ S,\ Q,\ \chi,\ \Pi,\ \sigma_{\mathrm{ext}})\,. \tag{48}$$

The quantity Q connects the inner binding structure with the range of the outer coupling. It therefore forms the central bridge between the atomic nucleus and electron shell, as well as between the galactic centre and the outer release field.

4.6. Preliminary Element–Galaxy Classification

The Phraqtal Field mapping does not connect an element with a galaxy through a single absolute value. A classification results from the interaction of several organisational axes. Each axis captures a different property of the bound overall system: the shape of its outer distribution, the position of its transition region, its response capability, its temporal hierarchy, and the share of its outer release organisation.

The complete working coordinate is

$$\mathcal{S} = (W,\ S,\ Q,\ \chi,\ \Pi,\ \sigma_{\mathrm{ext}})\,. \tag{49}$$

The individual components have the following functions:

$$
\begin{aligned}
W &: \text{normalised width of the outer distribution,} \\
S &: \text{concentration between inner and outer organisation,} \\
Q &: \text{relation between transition radius and outer radius,} \\
\chi &: \text{response or reorganisation capability,} \\
\Pi &: \text{slow precession or modulation relation,} \\
\sigma_{\text{ext}} &: \text{share of the outer release organisation.}
\end{aligned}
$$

These quantities are not used to create a rigid ranking. They form an organisational signature. Two systems belong to the same Phraqtal Field class if their signatures show the same distribution of inner binding, outer coupling, and dynamic reorganisation.

The profiles and transition relations determined so far initially produce three working classes.

Tabelle 10: Preliminary element–galaxy classes based on the profile and transition relations determined so far.

Phraqtal Field class	Atomic reference	Galactic candidate	dominant organisational features
I	H, Ne	UGC 01230	extended outer shell, broad coupling zone
II	Be, Mg, Ar	NGC 5033	multi-shell transition organisation, more compact outer binding
III	Ca	NGC 2955	more developed outer occupancy, strong reorganisation capability

4.6.1 Class I: Extended Outer Coupling Organisation

The first class is represented by hydrogen and neon. In the normalised profiles used so far, both show an extended outer organisational zone. UGC 01230 has a corresponding galactic signature: its outer dynamic and visible structure extends far beyond the inner baryonic core region.

Hydrogen represents the simplest bound shell organisation. Neon, within the same class, represents a more densely formed and stabilised outer occupancy. The classification of UGC 01230 in this class therefore does not describe uniformity, but a shared organisational family with extended outer coupling.

4.6.2 Class II: Multi-Shell Transition Organisation

The second class consists of beryllium, magnesium, and argon. Their shared property is a clearly developed relation between inner binding, outer shell structure, and the transition zone. NGC 5033 shows the strongest approximation to this class so far.

In particular, the transition relation of argon

$$Q_{\mathrm{Ar}} = 3.29 \tag{50}$$

is close to the relation determined for NGC 5033

$$Q_{\mathrm{NGC\,5033}} = 4.02. \tag{51}$$

NGC 5033 is therefore treated as an argon-like candidate in the current analysis. The galaxy combines a compact inner organisation with a clearly defined, but still dynamically coupled, outer structure. This class is particularly important for further study because profile shape, transition radius, and outer reorganisation capability can all be compared within it.

4.6.3 Class III: Extended Outer Occupancy

The third class is currently represented by calcium and NGC 2955. With its outer $4s^2$ occupancy, calcium has an extensive and highly reorganisable outer structure. Within the profile analysis so far, NGC 2955 shows a corresponding galactic class with a more developed outer coupling architecture.

This class describes systems whose stability does not follow solely from a compact centre. It arises from the balance between central binding, outer occupancy, and ongoing reorganisation. Further analysis still needs to determine the transition quotient Q_G, the outer release fraction σ_{ext}, and suitable precession quantities for NGC 2955.

The three classes do not form a closed periodic table of galaxies. They are the first organisational framework in which elements and galaxies are structured according to the same organisational roles. With additional data, further element families, galactic intermediate classes, and transition forms can be placed within the same Phraqtal Field space.

4.7. Central Binding, Core Capacity, and the Galactic Central Region

The outer occupancy of a bound system is limited and organised by its central binding organisation. In the atom, this central organisation appears as nuclear charge and atomic number Z. In the galactic regime, it appears as a compact central region consisting of baryonic concentration, inner dynamics, central release structure, and, where applicable, a saturated black-hole state.

The atomic number

$$Z = N_p \tag{52}$$

gives the number of protons in the atomic regime and therefore the central binding capacity of the neutral atom. It does not determine only the number of possible electrons. It also organises the shell structure, the energetic depth of the binding, and the reorganisation possibilities of the outer shell.

In the Phraqtal Field, Z is read as the atomic central index:

$$\mathcal{K}_{\mathrm{Zentrum},E} = \mathcal{K}(Z, \mathcal{A}_E). \tag{53}$$

Here, $\mathcal{K}_{\text{Zentrum},E}$ describes the ability of the atomic centre to carry a particular outer coupling architecture.

For galaxies, a corresponding central organisational index $\mathcal{K}_{\text{Zentrum},G}$ is introduced:

$$\mathcal{K}_{\text{Zentrum},G} = \mathcal{K}\left(\Sigma_{\text{bar}}, V_{\text{in}}, \mathcal{E}_{\text{rel}}, C_{\text{in}}\right). \tag{54}$$

Here, Σ_{bar} denotes the inner baryonic surface density, V_{in} the characteristic inner dynamics, $\mathcal{E}_{\text{rel}}$ the central release strength, and C_{in} the coherence of the inner bound structure. The specific normalisation of this index will only be established in a later quantitative analysis.

The central galactic organisation determines which outer coupling architecture can be stably supported:

$$\mathcal{K}_{\text{Zentrum},G} \quad \longrightarrow \quad \left(N_{\text{eff},G}, q_{\text{B}}, q_{\text{A}}, \sigma_{\text{ext}}\right). \tag{55}$$

A more concentrated central region can carry an outer structure organised differently from that of a diffuse central region. This does not mean that larger galaxies directly correspond to higher atomic numbers. What matters is the combination of inner binding, outer occupancy, transition scale, and release state.

4.7.1 Helium as the First Central Two-Particle Binding

Helium occupies a special position within this structure:

$$\text{He}: \qquad Z = 2, \qquad 1s^2. \tag{56}$$

With $Z = 2$, the nucleus carries a fully closed first shell for the first time. This creates the first form of matter in which central binding and outer two-particle occupancy form a stable balance. Helium is therefore the earliest reference for a closed coupling architecture.

The Phraqtal Field counterpart is a galaxy whose inner and outer organisations are not merely present, but already mutually stabilised:

$$\mathcal{K}_{\text{Zentrum},G} \leftrightarrow \mathcal{K}_{\text{auen},G}. \tag{57}$$

Such a class has a bound inner centre of mass and a coherent outer structure. This structure may be visibly compact, diffuse, or multi-level. What matters is that the outer organisation is not only temporarily bound to the centre, but has entered a stabilised overall state.

4.7.2 Central Saturation and Outer Release

In the Phraqtal Field, outer release does not arise independently of the centre. It is the response of the overall system to ongoing inner binding, radiation, saturation, and reorganisation. The working relation is

$$\Delta g_{\text{ent}}(r) = \mathcal{R}\left[\mathcal{K}_{\text{Zentrum},G}, \mathcal{B}_G(r), C_G(r)\right], \tag{58}$$

where $\mathcal{B}_G(r)$ denotes the baryonic organisation, $C_G(r)$ the radial coherence, and $\mathcal{R}$ the field response.

The additional outer acceleration is therefore understood as the reorganisation response of the overall galactic field state. It depends on the central region, but is distributed across the entire coupled galactic structure. This is precisely why galaxies with similar total baryonic mass can have different transition scales, peak or diffuse forms, and outer release fractions.

The extended Phraqtal Field signature is therefore

$$\mathcal{S}_{E,G} = \left(\mathcal{K}_{\text{Zentrum}}, \mathcal{A}, W, S, Q, \chi, \Pi, q_{\text{B}}, q_{\text{A}}, T, \sigma_{\text{ext}}, N_{\text{eff}}\right). \tag{59}$$

It connects central binding, outer occupancy, transition structure, reorganisation capability, and release state in a common organisational description. The classification of an element into a galaxy class is therefore based on the complete signature and not on mass, size, or a single visible property.

4.8. Element–Galaxy Mapping as an Organisational Matrix

In the Phraqtal Field, the mapping of elements and galaxies is not based on a direct equality of size, mass, or particle number. It is based on recurring organisational roles: central binding, outer coupling capability, stabilisation, reorganisation, and release.

A chemical element forms a stable field organisation bound on the atomic scale. A galaxy forms a stable field organisation bound on the galactic scale. In both cases, the relationship between central binding and outer coupling capability determines how many states are permanently integrated, how easily new states are taken in, and under which conditions reorganisation begins.

For an element E, the organisational state description is defined by

$$\mathcal{F}_E = (K_E,\ O_E,\ R_E,\ C_E)$$

where K_E denotes the central binding and closure strength, O_E the outer opening or coupling capability, R_E the reorganisation readiness, and C_E the inner coherence.

The corresponding galactic description is

$$\mathcal{F}_G = (K_G,\ O_G,\ R_G,\ C_G).$$

Here, K_G describes the stabilisation provided by the central region, the inner baryonic system, and the bound overall organisation. O_G describes the coupling capability of the outer region: gas, disk, halo, satellite systems, and the connection to the surrounding field. R_G describes the tendency toward transitions, switching, and reorganisation. C_G describes the coherence with which this organisation becomes visible and dynamically effective in our observation regime.

The atomic and galactic organisational roles are therefore given by

$$\mathcal{F}_E \longleftrightarrow \mathcal{F}_G$$

4.8.1 Hydrogen: Open Initial Organisation

Hydrogen is the first stable open matter organisation. The core region has minimal central binding, while the outer electron organisation has a high capability for coupling, forming bonds, and reorganisation. Hydrogen is therefore not defined by maximum closure, but by productive openness.

$$K_{\mathrm{H}} \text{ low to medium}, \qquad O_{\mathrm{H}} \text{ high}, \qquad R_{\mathrm{H}} \text{ high}.$$

The galactic hydrogen family includes open, gas-rich, and matter-building systems. Its outer organisation can take in new couplings, condense gas, form young stellar populations, and integrate satellite structures into the host organisation. The high outer openness allows pronounced transition zones between the inner centre and the galactic outer region.

This class is particularly relevant for galaxies in which formation zones, gas accumulations, and visible young stellar populations occur together. Visibility is a consequence of the coherence achieved, not a prerequisite for the underlying field structure to already exist.

4.8.2 Helium: First Closed Stability Organisation

Helium forms the first closed two-particle organisation. The first shell is fully integrated, creating a state of high coherence and low outer reorganisation capability. In the Phraqtal Field, helium is the first common form of matter in which binding, saturation, and stabilisation act together.

$$K_{\mathrm{He}} \text{ high}, \qquad O_{\mathrm{He}} \text{ low}, \qquad R_{\mathrm{He}} \text{ low}, \qquad C_{\mathrm{He}} \text{ high}.$$

The corresponding galactic helium family describes systems with strong inner integration. Their release fields continue to operate, but are distributed across the overall organisation and are not necessarily tied to a single strong transition zone. Outer coupling capability remains present, but it is limited and organised by the high inner coherence.

Helium therefore represents galaxy families whose structure remains stably integrated over long periods. Their development takes place mainly through slow redistribution, the gradual uptake of outer structures, and coherent adjustment of the overall field.

4.8.3 Beryllium and Magnesium: Renewed Outer Opening

With beryllium and magnesium, an outer binding capability appears again after a stabilised inner organisation. These elements combine a closed core region with a comparatively open outer shell. Their organisational role is therefore defined by the simultaneous presence of stability and reorganisation capability.

$$K_E \text{ high}, \qquad O_E \text{ increased again}, \qquad R_E \text{ medium to high}.$$

On the galactic scale, this family corresponds to systems whose central organisation is stably

developed, while their outer region can couple new matter, gas structures, or satellites. These galaxies have a clear inner order and, at the same time, an active transition region between the centre, disk, and outer field.

They form an important class for studying the Q-switch: the inner structure is not dissolved, but the outer organisation can enter an activated reorganisation state.

4.8.4 Neon and Argon: Closed Multi-Shell Organisation

Neon and argon represent closed multi-shell organisations. Their outer binding is strongly saturated with respect to additional couplings. The structural role of these elements consists in a high degree of self-integration: the existing system carries its organisation largely from within itself.

$$K_E \text{ high}, \qquad O_E \text{ low}, \qquad R_E \text{ low}, \qquad C_E \text{ high}.$$

The corresponding galactic family has a largely integrated outer field response. Release does not occur only at individual points, but across a spatially connected organisation. Such systems can display a diffuse acceleration structure without this indicating a lack of dynamic activity. In this case, the activity is evenly integrated into the overall structure.

4.8.5 Calcium: Strong Outer Coupling and Integration Capability

Calcium represents an organisation with a stable inner structure and a clearly available outer binding capability at the same time. This combination allows pronounced interaction with other systems without losing central cohesion.

$$K_{\mathrm{Ca}} \text{ high}, \qquad O_{\mathrm{Ca}} \text{ high}, \qquad R_{\mathrm{Ca}} \text{ high}.$$

The galactic calcium family describes large systems that are strongly bound at the centre and have a continuously active outer organisation. They can take in, rearrange, and integrate gas, satellites, and outer field structures over long periods. Their development is not determined by a single switching event, but by an ongoing sequence of coupling, release, reorganisation, and renewed stabilisation.

The matrix does not define a closed list of individual element–galaxy pairs. It creates an organisational space in which each specific galaxy can be assigned to an element family based on its radial field shape, its activation state, its gas organisation, its satellite coupling, and its visibility.

4.9. Overall Matrix of Atomic and Galactic Organisational Roles

In the Phraqtal Field, atomic and galactic organisation are compared through recurring roles. The mapping does not concern absolute sizes, orbits, or individual object counts. It concerns the function of a structure within a bound overall organisation.

Tabelle 11: Preliminary organisational matrix of the element–galaxy mapping in the Phraqtal Field.

Element family	Atomic organisational role	Galactic organisational role	Dominant developmental mode
Hydrogen	open first binding; high outer coupling	gas-rich, building, or locally strongly reorganising galaxy	condensation, star formation, new coupling
Helium	first closed, coherent two-particle binding	long-term integrated stability galaxy	distributed release, slow rearrangement
Beryllium/ Magnesium	stable inner region with renewed open outer shell	stable galaxy with active transition zone	Q-activation, uptake, and reorganisation
Neon/Argon	closed multi-shell organisation	strongly integrated, broadly ordered galaxy	diffuse self-stabilisation
Calcium	strong core with available outer binding capability	large, centrally bound and externally coupled galaxy	ongoing integration of outer structures

4.9.1 The Central Correspondence

The core region of an atom and the central region of a galaxy are both locations of maximum binding density within their respective organisations. They carry the inner stability of the overall system and determine how many outer coupling states can be permanently integrated.

$$K_{\text{Atom}} \longleftrightarrow K_{\text{Galaxie}}.$$

During a strongly integrated phase, the central region stabilises the entire organisation. When the inner tension reaches a saturation threshold, the same region becomes the starting point for release and reorganisation.

4.9.2 The Outer Correspondence

Electrons form the outer binding organisation of an atom. They are not classically orbiting individual bodies, but bound field modes with different coupling and reorganisation possibilities.

On the galactic scale, satellites, gas structures, and outer stellar populations take on the corresponding role. They are partial organisations of the host galaxy, can be coupled with different strengths, and may become visible, separate, or be integrated during their development.

$$\text{electron organisation} \longleftrightarrow \text{satellite and outer organisation.}$$

Tabelle 12: Overall matrix of atomic and galactic organisational roles in the Phraqtal Field.

Organisational role	Atomic form	Galactic form
Central binding	core region as a highly bound central matter organisation	central condensation and saturation region of the host galaxy
Outer binding	electrons as bound outer field modes	gas, stellar populations, satellites, flows, and outer field organisation
Occupancy	stable number and arrangement of outer electron modes	effective number and coupling of outer partial organisations
Open organisation	available outer binding capability; readiness for chemical reorganisation	gas uptake, star formation, satellite coupling, and active transition zones
Closed organisation	fully integrated shell; high inner coherence	broadly stabilised galaxy with long-term integrated outer organisation
Saturation	excited, ionised, or unstable central state	highly saturated central region, active quasar, or strong release phase
Release	spectral line, particle emission, or reorganisation into a more stable state	radiation, outflow, jet, gas reorganisation, and outer matter-forming effect
Matter formation	binding of minimal field modes into new stable matter states	pair organisation, gas condensation, star formation, and formation of new partial structures
Integration	change of a bound outer electron state	merging, incorporation, or redistribution of satellites and gas
Visibility	spectrum as a coupling-capable modulation of a bound state	optical, infrared, radio, or X-ray-active field readability

In both cases, the relevant quantity is not the simple number, but the stable effective occupancy:

$$N_{\text{eff}} = \sum_i \kappa_i C_i \Xi_i .$$

4.9.3 The Saturation Correspondence

A heavy, excited, or unstable atomic state carries increased central reorganisation tension. It releases this tension through radiation, particle emission, or transitions into new bound states.

A quasar or highly active central region carries a corresponding saturation organisation on the galactic scale. The release appears as strong radiation and as an effect on gas, star formation, and the outer field structure:

$$\text{atomic release} \longleftrightarrow \text{quasar release}.$$

The quasar phase extends this role through its matter-formation and feedback function. Central inflow, release, outer matter organisation, and return flow together form a cyclical overall state.

4.9.4 The Visibility Correspondence

A spectrum makes an atomic state transition readable. It does not show the entire bound organisation, but rather the modulation that couples to the observer.

Galactic visibility works in the same way. Starlight, gas emission, radio sources, X-ray radiation, or dynamic acceleration profiles make different aspects of the same galactic organisation readable:

$$\text{atomic spectral line} \longleftrightarrow \text{galactic signature}.$$

The overall matrix therefore connects atom and galaxy through a shared logic of binding, outer occupancy, saturation, release, matter formation, integration, and visibility. The Phraqtal Field makes this logic readable as a recurring organisational grammar of the Phraqtal Field.

5. Bimodality and Q-Activation

5.1. Bimodal Field Forms and Dynamic Activation

The previous studies of galactic acceleration profiles provide two complementary differentiation axes for the Phraqtal Field [7, 8].

The first axis describes the shape of the outer release organisation. It is captured by the bimodal regime structure. Here, the corresponding shape scale is denoted by q_{B}:

$$q_{\text{B}} = \text{bimodal shape scale of the outer field response.} \tag{60}$$

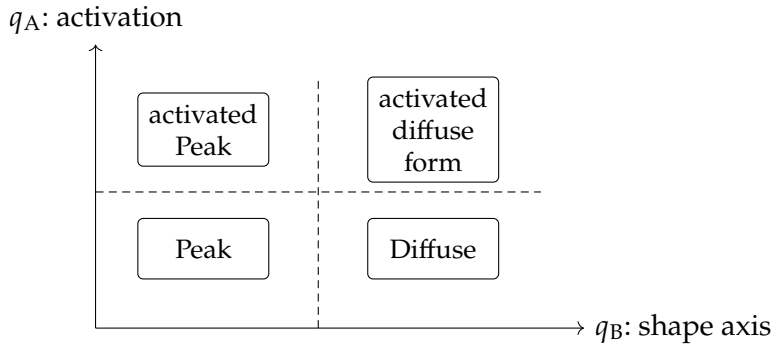


Abbildung 5: The shape axis q_B and the activation axis q_A are separate differentiation axes.

The second axis describes whether a finite transition scale has been dynamically switched on. It is captured by the activation analysis and denoted by q_A:

$$q_A = \text{dynamically activated transition scale.} \tag{61}$$

The distinction is necessary because both quantities arise from different fitting and normalisation procedures. q_B classifies the shape of the field response; q_A describes its dynamic activation state.

The following threshold ranges and activation limits are adopted from the profile analyses described above [8].

For the bimodal axis, the three empirical ranges are

$$\begin{aligned} q_B &< 1.2 \quad &&\text{: Peak regime,} \\ 1.2 \le q_B &< 2.5 \quad &&\text{: transition regime,} \\ q_B &> 2.5 \quad &&\text{: diffuse regime.} \end{aligned} \tag{62}$$

The Peak regime describes a locally concentrated release organisation with a pronounced maximum. It occurs mainly in compact, small, faint, or dwarf-galaxy systems. The diffuse regime describes an outer field response distributed over large radii, as more often seen in more massive spiral galaxies. Between the two lies a smaller transition population.

The activation axis is determined by the turnover strength T of the acceleration profile:

$$T = s_{\text{in}} - s_{\text{out}}, \tag{63}$$

where s_{in} and s_{out} denote the inner and outer logarithmic profile slopes. The activation threshold is

$$T_c \approx 4.3. \tag{64}$$

This gives three activation states:

$$\begin{aligned} T < T_c - \Delta T \quad &\Rightarrow \quad \text{not activated,} \\ |T - T_c| \le \Delta T \quad &\Rightarrow \quad \text{transition state,} \\ T > T_c + \Delta T \quad &\Rightarrow \quad \text{activated.} \end{aligned} \tag{65}$$

In the non-activated state,

$$q_{\mathrm{A}} \approx 0. \tag{66}$$

The outer field response is present, but does not yet have a developed dynamic transition scale. In the activated state, a finite scale emerges:

$$q_{\mathrm{A}} > 0. \tag{67}$$

It marks a large-scale reorganisation of the acceleration structure.

The Phraqtal Field classification is therefore extended to

$$\mathcal{F} = \left(W,\, S,\, Q,\, \chi,\, \Pi,\, \sigma_{\mathrm{ext}},\, q_{\mathrm{B}},\, q_{\mathrm{A}},\, T\right). \tag{68}$$

The first six quantities describe the bound organisational signature. q_{B} determines the shape class of the outer release. q_{A} and T determine whether this form has already developed a dynamically activated transition scale.

This creates not merely three galactic classes, but a two-axis Phraqtal Field space:

Tabelle 13: Two-axis differentiation of galactic Phraqtal Field states.

Shape class	Activation state	Phraqtal Field interpretation
Peak	not activated	local outer organisation that has not yet been fully switched
Peak	activated	concentrated release with a developed transition scale
Transition	near T_c	reorganisation between local and distributed field forms
Diffuse	not activated	large-scale, weakly switched outer organisation
Diffuse	activated	large-scale coherent release with a finite transition scale

Bimodality and the Q-switch therefore complement each other: bimodality determines *how* the outer field response is organised spatially; the switch determines *whether* the corresponding transition organisation has been dynamically activated. Only both axes together allow a more precise mapping between atomic binding classes, galactic release forms, and visibility states.

5.2. Two Q-Axes of Phraqtal Field Differentiation

Galactic bimodality and the dynamic switch describe two different properties of the same field organisation. Only their joint consideration allows a differentiated assignment of element families and galaxy families in the Phraqtal Field.

The first axis describes the *spatial shape* of the outer field response. It is denoted by q_{B}. The subscript B stands for *bimodal*. Small values correspond to a concentrated field response that is strongly pronounced within a limited radial range. Large values correspond to a broadly distributed, diffuse outer response. Thus, q_{B} does not distinguish between visible and invisible, but between different forms of organisation.

The second axis describes the *dynamic activation state* of a transition scale. It is denoted by q_{A}. The subscript A stands for *activation*. q_{A} remains close to zero as long as the inner and outer organisations are coupled without a pronounced dynamic switch. When the transition tension exceeds a critical threshold, a finite transition scale is activated. A suitable working form is

$$q_{\mathrm{A}} = q_{\max}\left[1 + \exp\left(-\frac{\mathcal{T} - \mathcal{T}_{\mathrm{c}}}{w}\right)\right]^{-1},$$

where $\mathcal{T}$ denotes the strength of the inner–outer transition, $\mathcal{T}_{\mathrm{c}}$ the activation threshold, and w the width of the switching range.

Each galaxy in the Phraqtal Field therefore has at least two independent organisational coordinates:

$$\mathcal{G} = \left(q_{\mathrm{B}}, q_{\mathrm{A}}\right).$$

q_{B} determines *how* the outer field response is distributed spatially. q_{A} determines *whether and how strongly* a dynamic transition scale between inner binding and outer reorganisation is activated.

5.2.1 Bimodal Shape Classes

The shape classification derived from rotation curves can be written in the Phraqtal Field as follows:

$$\begin{aligned} q_{\mathrm{B}} < 1.2 \quad &: \text{ Peak organisation,} \\ 1.2 \le q_{\mathrm{B}} < 2.5 \quad &: \text{ transition organisation,} \\ q_{\mathrm{B}} \ge 2.5 \quad &: \text{ diffuse organisation.} \end{aligned}$$

A Peak organisation has a comparatively concentrated outer field response. Structurally, it is connected to a local reorganisation zone in which the coupling between the central region, disk, gas accumulations, and outer field profile appears clearly concentrated. Such states are especially expected in small, gas-rich, and structurally open galaxies.

A diffuse organisation, by contrast, has no single dominant outer reorganisation zone. Its release is distributed across a large radial range. This class fits galaxies whose outer field response is broad, continuous, and integrated over longer periods.

The transition class connects both forms. It contains systems in which the outer organisation either becomes more concentrated or broader, without the transition already being fully complete.

5.2.2 Dynamic Activation Classes

Regardless of the shape class, q_{A} describes the dynamic state:

$$\begin{aligned} q_{\mathrm{A}} \simeq 0 \quad &: \text{ non-activated state,} \\ 0 < q_{\mathrm{A}} < q_{\max} \quad &: \text{ transition state,} \\ q_{\mathrm{A}} \to q_{\max} \quad &: \text{ activated state.} \end{aligned}$$

In the non-activated state, field tension is carried by the existing inner and outer organisation. In the transition state, the difference between inner and outer reorganisation behaviour increases. The activated state contains a clearly developed transition scale at which the field response changes its character.

Activation is therefore not an additional matter component. It is the dynamic state of the Phraqtal Field through which the observed acceleration structure is organised radially.

5.2.3 Atomic Correspondence

At the atomic level, the two axes do not correspond to individual electron orbits, but to two properties of the bound electron organisation. The atomic shape axis describes the spatial distribution of electron density between the region near the nucleus and the outer binding region. The atomic activation axis describes whether an outer binding shell is closed, open, or reorganised in a transition state.

For an element E we therefore write

$$\mathcal{E} = (b_E, a_E)\,,$$

where b_E denotes the radial shape of the bound electron organisation and a_E its coupling and activation state. The mapping is based on the organisational role:

$$(b_E, a_E) \longleftrightarrow \left(q_{\mathrm{B}}, q_{\mathrm{A}}\right).$$

Hydrogen forms the open initial class. Its single bound electron organisation represents the simplest stable coupling between the core region and the outer field. Helium, by contrast, forms the first closed and particularly stable two-particle organisation. With the first shell fully occupied, an early state of high inner coherence and low outer reorganisation readiness emerges.

In the Phraqtal Field, helium therefore corresponds to a galactic stability family in which the central and outer organisations remain coupled in a closed form over long periods. This family need not be field-inactive: release also takes place there. However, it is broadly distributed, repeatedly integrated, and less characterised by abrupt structural switches.

5.2.4 Four Basic Galactic States

The combination of both Q-axes produces four particularly important basic galactic states:

	$q_{\mathrm{A}} \simeq 0$	$q_{\mathrm{A}} > 0$
small q_{B}	concentrated stable	locally activated
large q_{B}	diffuse stable	broadly activated

The concentrated stable state has a locally concentrated outer organisation without a strong dynamic switch being activated. The locally activated state, on the other hand, contains a clear reorganisation zone and is especially relevant for gas condensation, satellite coupling,

Tabelle 14: Working mapping of the early atomic and galactic organisational families in the Phraqtal Field.

Atomic family	Binding and coupling state	Galactic counterpart
Hydrogen	open first binding organisation; high capability for reorganisation and new coupling	open, gas-rich, or developing galaxy organisation; local reorganisation zones are possible
Helium	first closed two-particle organisation; high inner coherence and stable saturation	integrated stability family; broadly distributed release, lower tendency toward abrupt switches
Elements with a renewed open outer shell	additional outer binding capability; transitions and reorganisations of the outer structure	galaxies with a pronounced transition scale, satellite coupling, or active redistribution between inner and outer regions

and the formation of young visible stellar populations.

The diffuse stable state corresponds to a largely integrated galactic organisation. Release fields are present, but their radiation, strength, and visibility depend on coherence, geometry, and coupling to the observing regime. The broadly activated state finally describes systems in which large-scale reorganisation takes place across an extended region of the galaxy.

The element–galaxy mapping is therefore not determined by the simple number of electrons or satellites. What matters is the shared position of a system in the two-dimensional organisational space formed by shape, activation, coherence, and coupling capability.

5.3. Operational Mapping Rule for Atomic and Galactic Switching States

Bimodality and the dynamic Q-switch allow a more precise element–galaxy mapping when the same organisational questions are asked in both regimes:

1. Is the outer organisation locally concentrated or diffusely distributed?
2. Does the system have a clearly developed transition zone?
3. Is this transition zone stably bound, being rebuilt, or dynamically activated?
4. Is the outer organisation open, closed, or able to take on further occupancy?

The first question is answered by the shape class. The second and third questions are answered by the transition relation and the activation state. The fourth question connects the atomic electron occupancy with the galactic outer occupancy made up of satellites, gas, star-formation zones, and release structures.

5.3.1 Atomic Switching States

For the atomic level, a discrete occupancy state $\mathcal{A}_E$ is introduced:

$$\mathcal{A}_E \in \{\text{open, closed, further occupiable}\}\,. \tag{69}$$

Hydrogen has the first open matter organisation:

$$\mathrm{H}: \qquad 1s^1, \qquad \mathcal{A}_{\mathrm{H}} = \text{open}. \tag{70}$$

Its outer structure is bound, but not yet closed as a complete first shell. In the Phraqtal Field, hydrogen therefore represents the first simply bound matter class that remains directly capable of reorganisation.

Helium has the first closed matter organisation:

$$\mathrm{He}: \qquad 1s^2, \qquad \mathcal{A}_{\mathrm{He}} = \text{closed}. \tag{71}$$

The first shell is fully occupied. Helium therefore represents the first stable closure of a bound outer organisation. It does not represent merely a subsequent state of hydrogen, but the first complete stabilisation of a two-particle coupling mode.

Elements such as beryllium, magnesium, and calcium again possess outer states that can be occupied further beyond closed inner shells:

$$\mathcal{A}_E = \text{further occupiable}. \tag{72}$$

They form the class of multi-shell and outward-open matter organisations.

Tabelle 15: Atomic occupancy states as early Phraqtal Field classes.

Atomic class	Reference	Outer structure	$\mathcal{A}_E$
Simply open	H, $1s^1$	first bound single coupling	open
First closure	He, $1s^2$	complete first two-particle binding	closed
Open multi-shell	Be, Mg, Ca	outer shell beyond inner closure	further occupiable
Closed multi-shell	Ne, Ar	completed outer shell organisation	closed

5.3.2 Galactic Switching States

The galactic counterpart is described by two separate coordinates. The first coordinate q_{B} determines the spatial shape of the release organisation:

$$\mathcal{B}_G \in \{\text{Peak, transition, diffuse}\}. \tag{73}$$

The second coordinate describes the activation state of the transition scale:

$$\mathcal{A}_G \in \{\text{not activated, transition, activated}\}. \tag{74}$$

The galactic state class is therefore written as

$$\mathcal{G} = \left(\mathcal{B}_G,\, \mathcal{A}_G,\, N_{\mathrm{eff},G}\right). \tag{75}$$

$N_{\text{eff},G}$ denotes the effective outer occupancy of the host galaxy. It includes not only directly visible dwarf galaxies, but all sufficiently bound outer partial organisations. Visible satellites, gas accumulations, and young stellar populations are different observational forms of the same outer occupancy architecture.

5.3.3 Mapping as a State Matrix

Atomic and galactic classification is not equated through the absolute number of electrons or satellites. The mapping is based on the shared organisational role of outer occupancy and transition behaviour.

Tabelle 16: Phraqtal Field matrix for the element–galaxy mapping.

Atomic organisational role	Galactic organisational role	Phraqtal Field interpretation
open single coupling	open, far-reaching outer coupling	early or weakly integrated structure
first closed two-particle binding	stabilised inner and outer coupling	early stable overall organisation
multi-shell, further occupiable	multi-level satellite and gas occupancy	outer structure that can be further developed
multi-shell closed	integrated, coherent outer organisation	strongly stabilised overall architecture

The bimodality and activation quantities further differentiate this matrix. A galaxy with an open outer occupancy can, for example, have a Peak profile with an activated transition scale. Another open outer occupancy can be diffuse and not yet activated. The element class alone is therefore not enough; it is only fully determined by the galactic shape class and the switching state.

The complete mapping rule is therefore

$$\mathfrak{M}_{E,G} = \left(\mathcal{A}_E,\, W,\, S,\, Q,\, \chi,\, \Pi;\, \mathcal{B}_G,\, \mathcal{A}_G,\, \sigma_{\text{ext}},\, N_{\text{eff},G}\right). \tag{76}$$

For the next quantitative analysis, the atomic reference profiles will be examined using the same normalised profile analysis that has already been used for the galaxies. This makes it possible to determine an atomic shape class

$$\mathcal{B}_E \in \{\text{Peak, transition, diffuse}\} \tag{77}$$

for each element. Only then will it be tested which galactic Peak, transition, and diffuse classes correspond to the respective atomic binding states.

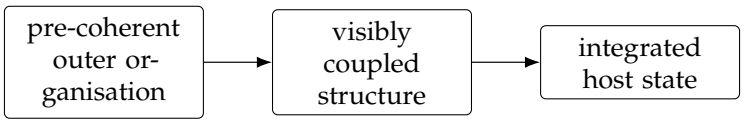


Abbildung 6: Outer nodes can become visible, be reorganised, and return to the host organisation.

6. Reorganisation, Satellites, and Outer Coupling

6.1. Reorganisation Cycle, Visibility, and Integration

In the Phraqtal Field, bound structures are not static objects. They continuously pass through state changes between formation, coherence building, visible coupling, and integration. This cycle occurs in all regimes. Its temporal resolution and visible form differ, while the organisational sequence remains the same.

The general reorganisation cycle is

$$\mathcal{Z} = (\mathcal{E}, \mathcal{K}, \mathcal{V}, \mathcal{I}), \tag{78}$$

where

$\mathcal{E}$: formation or release state,
$\mathcal{K}$: coherence building and bound stabilisation,
$\mathcal{V}$: visible coupling to the observation regime,
$\mathcal{I}$: integration or reorganisation in the overall system.

6.1.1 Atomic Reorganisation

In the atomic regime, bound matter organisation begins with coupling-capable electron–positron structures. Under conditions of field saturation and release, local pair states initially form. From these, more stable particle organisations develop, from which proton–neutron bonds and the first bound elements emerge.

Hydrogen represents the first open bound form of matter:

$$\mathcal{E} \longrightarrow \mathrm{H}(1s^1). \tag{79}$$

Helium forms the first stable closure:

$$2\,\mathrm{H} \longrightarrow \mathrm{He}(1s^2). \tag{80}$$

Compared with hydrogen, the closed helium structure has greater internal stabilisation. In the Phraqtal Field, it marks the first transition from an open single binding to a complete two-particle coupling. This creates an early matter class with lower outer reorganisation capability and stronger structural retention.

Atomic excitation, ionisation, recombination, and emission are state changes of this bound

overall system. They do not abolish the atomic organisation, but temporarily rearrange its coupling structure:

$$\text{bound} \longrightarrow \text{excited or open} \longrightarrow \text{reorganised bound}. \tag{81}$$

6.1.2 Galactic Reorganisation

In the galactic regime, the corresponding cycle begins in release and matter-formation zones. Gas accumulations, condensed outer nodes, and young satellite structures initially emerge as bound pre-organisations. Their complete visibility does not yet depend solely on their existence, but on the coherence they have achieved and their coupling to our observation regime.

For a galactic partial structure i, visibility is described by

$$\mathcal{V}_i = C_i \, \mathcal{K}_i \, \mathcal{R}_i \tag{82}$$

where C_i denotes the coupling strength, $\mathcal{K}_i$ the inner coherence, and $\mathcal{R}_i$ the transferability of its radiation or release signature into the observation regime.

A structure becomes visible when

$$\mathcal{V}_i \geq \mathcal{V}_{\text{krit}}. \tag{83}$$

Below this threshold, it is not without structure. It is in a pre-visible or weakly visible coupling state.

The galactic developmental sequence is therefore

$$\begin{aligned} &\text{release zone} \\ &\longrightarrow \text{gas and pre-structure} \longrightarrow \text{young stellar population} \\ &\longrightarrow \text{visible satellite or outer node} \\ &\longrightarrow \text{integration into the host galaxy}. \end{aligned} \tag{84}$$

The formation of a young stellar population increases $\mathcal{R}_i$ and thus the visible regime coupling. A gas or satellite structure that was previously only weakly visible or invisible can thereby appear as an independent dwarf galaxy, star-formation region, or outer galactic node.

6.1.3 Integration as Bound Reorganisation

The integration of a satellite structure is not the destruction of its organisation. Its bound structure is rearranged through long-term coupling with the host field. Gas, stars, orbital momentum, twist, and release fractions gradually pass into the overall organisation of the host galaxy:

$$\mathcal{I}_G : \qquad \mathcal{S}_{\text{Satellit}} + \mathcal{S}_{\text{Wirt}} \longrightarrow \mathcal{S}'_{\text{Wirt}}. \tag{85}$$

In the atomic regime, this corresponds to the rearrangement of bound electron modes. An electron state is not pulled into the nucleus as a classical body, but transferred into another bound, excited, or open coupling configuration. On the galactic level, a satellite is not simply

erased; its organisation is integrated into the larger bound host structure.

6.1.4 Connection to the Q-Switch

The Q-switch determines whether a system carries a clearly developed transition scale. In the reorganisation cycle, this corresponds to the transition from an existing but not yet fully developed outer structure to a dynamically coherent release organisation:

$$q_{\mathrm{A}} \approx 0 \quad \longrightarrow \quad q_{\mathrm{A}} > 0. \tag{86}$$

An activated galaxy therefore does not necessarily possess more visible matter than a non-activated galaxy. It has a more developed dynamic transition organisation between inner baryonic binding and the outer release field. Bimodality determines the spatial shape of this organisation; the Q-switch determines its dynamic state of maturity.

The observed change between visibility, weak coupling, and integration is thus a direct consequence of the ongoing reorganisation cycle. The Phraqtal Field describes matter, radiation, and galactic dynamics as different visible states of the same bound field process.

6.2. Satellite Occupancy, Outer Electron Organisation, and Integration

The outer organisation of an atom and the outer organisation of a galaxy follow the same basic principle in the Phraqtal Field: a centrally bound core region does not have an unlimited ability to carry additional bound states permanently. The possible occupancy is determined by the coupling capability of the overall system.

In the atom, this occupancy appears as a stable outer electron organisation. In the galaxy, it appears as a system of gas accumulations, stellar populations, satellite galaxies, flows, and outer field structures. Satellites are not passive objects at the edge, but independent bound partial organisations whose lasting connection with the host galaxy depends on their mutual field compatibility.

Galactic occupancy can be written as a sum of effective outer coupling states:

$$N_{\mathrm{eff}}^{(G)} = \sum_{i=1}^{N_{\mathrm{sat}}} \kappa_i \, C_i \, \Xi_i .$$

Here, κ_i denotes the coupling strength of a satellite to the host galaxy, C_i its inner coherence, and Ξ_i the geometric and dynamic compatibility with the outer field of the host organisation. A satellite contributes permanently to the occupancy only when these quantities are sufficiently large together.

The atomic counterpart is

$$N_{\mathrm{eff}}^{(E)} = \sum_{j=1}^{N_e} \kappa_j^{(E)} \, C_j^{(E)} \, \Xi_j^{(E)} .$$

An electron is not defined by a classical orbit, but by a stable bound field mode. Its assignment to an atom arises when its outer organisation remains coherently coupled to the core region

and the states already bound there.

6.2.1 Formation of Outer Partial Organisations

Outer galactic partial organisations initially arise as field and matter condensations. Gas accumulations can form in such regions, condense, and develop their own bound structure. Only when this structure is sufficiently coherent and young stellar populations form does it couple visibly to our observation regime.

The visible satellite galaxy is therefore not the beginning of its organisation. It is the state in which the already existing partial organisation produces a measurable signal:

$$\mathcal{V}_i = C_i \, \kappa_i \, \mathcal{E}_i .$$

Here, $\mathcal{V}_i$ is the visibility, C_i the coherence, κ_i the coupling to our regime, and $\mathcal{E}_i$ the available release or emission strength.

If $\mathcal{V}_i$ is small, the partial organisation can exist without being clearly visible optically or radiatively. If $\mathcal{V}_i$ increases through growing coherence, star formation, or a stronger release organisation, it appears in our regime as a visible dwarf galaxy, gas structure, or young stellar population.

6.2.2 Coupling Levels

The outer occupancy of a galaxy is not one-dimensional. A satellite can couple to different levels of the host organisation: the central region, the rotating disk, the gas region, the outer acceleration profile, or more far-reaching regime couplings. These levels do not have to be active at the same time.

$$\mathcal{K}_i = \left(\kappa_{i,\text{Zentrum}}, \kappa_{i,\text{Scheibe}}, \kappa_{i,\text{Gas}}, \kappa_{i,\text{Auenfeld}}, \kappa_{i,\text{Regime}} \right) .$$

Strong coupling to the outer region combined with weak coupling to the disk produces a long-lived, spatially separate satellite state. Increasing coupling to the disk and centre, on the other hand, leads to gradual orbital change, material exchange, tidal deformation, and eventually integration into the host galaxy.

Integration is therefore not the disappearance of the organisation, but a redistribution of its coherence:

$$\text{separate satellite state} \longrightarrow \text{transition coupling} \longrightarrow \text{integrated host state.}$$

On the atomic scale, this corresponds to the reorganisation of an outer bound mode. An electron can change its binding state, move into another spatial configuration, or be transferred between bound organisational states. The basic organisation remains intact, while its visible and energetic expression changes.

6.2.3 Occupancy Limit and Stability

A host galaxy can stably couple outer partial organisations only up to a system-dependent limit. This limit is not a fixed number, but results from the relationship between inner binding strength, outer openness, and the current reorganisation tension:

$$\mathcal{B}_G = \frac{K_G\, C_G}{O_G + R_G}.$$

With high central binding and high coherence, a galaxy can organise and carry several outer partial organisations. If outer openness or reorganisation tension becomes too strong, individual satellites become more weakly coupled, are rearranged, or are integrated into the host organisation.

The corresponding atomic occupancy limit explains why not every additional electron organisation can be coupled arbitrarily to a core region:

$$\mathcal{B}_E = \frac{K_E\, C_E}{O_E + R_E}.$$

The maximum stable electron occupancy and the maximum stable satellite occupancy are therefore two scale-dependent expressions of the same organisational condition: stability arises from a balance between inner binding and outer openness.

6.2.4 Implications for Emerging and Disappearing Structures

The observed emergence or disappearance of an outer galactic structure describes a change of state in the Phraqtal Field. A partial organisation can pass through

$$\text{pre-coherent} \longrightarrow \text{visibly coupled} \longrightarrow \text{integrated}$$

In the first phase, its structure already exists, but has not yet achieved sufficient visibility in our regime. In the second phase, coherence, gas condensation, and star formation produce a measurable signal. In the third phase, its organisation increasingly becomes part of the host galaxy.

Satellites, gas structures, and young stellar populations therefore form a dynamic outer occupancy of the galaxy. Their number alone is not decisive. What matters is which partial organisations are coherently coupled, visibly activated, only weakly bound, or already integrated into the host organisation.

7. Saturation, Quasars, and Matter Return

7.1. Highly Saturated Central Organisations: Quasars and Atomic Release Classes

The previous element–galaxy matrix mainly describes stable, open, and closed organisational families. Quasars and very massive galaxies also form their own saturation class. Their

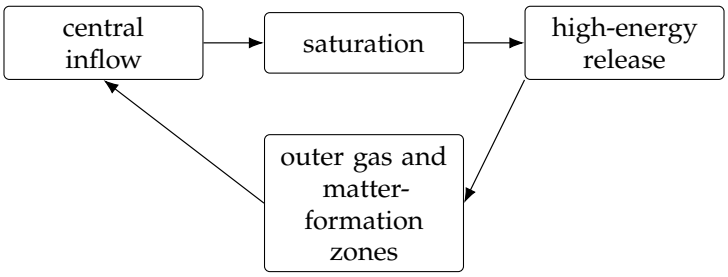


Abbildung 7: Quasar cycle as a sequence of saturation, release, outer reorganisation, and feedback.

essential feature is not simply their size or mass, but the relationship between centrally stored field tension and the ability of the system to release this tension in an organised way to the outside.

For a bound system X, we define the saturation state by

$$\Sigma_X = \frac{\mathcal{T}_{\text{zentral},X}}{\mathcal{E}_{\text{rel},X}},$$

where $\mathcal{T}_{\text{zentral},X}$ denotes the central field tension and $\mathcal{E}_{\text{rel},X}$ the currently available release capacity.

A system with a smaller Σ_X can continuously integrate its inner tension into the existing overall organisation. As Σ_X increases, release becomes concentrated. When saturation exceeds the regular integration capability, an active release state emerges:

$$\Sigma_X > \Sigma_{\text{krit}} \quad \Longrightarrow \quad \text{centrally concentrated reorganisation.}$$

7.1.1 The Atomic Saturation Class

On the atomic scale, high saturation does not appear as an ordinary closed electron shell. It appears in organisations whose core region or inner electron structure carries increased reorganisation tension. This includes, in particular, heavy, unstable nuclear organisations, excited states, and strongly ionised atoms.

The atomic saturation class has three interconnected forms:

Atomic state	Organisational role in the Phraqtal Field
Heavy stable nucleus	high central binding with sufficient internal integration capability; release remains largely organised
Excited or ionised state	the outer and inner electron organisation is not fully integrated; visible radiation emerges through reorganisation into a more coherent state
Heavy unstable nucleus	central saturation exceeds long-term integration capability; release appears as radioactive decay, particle emission, and energetic rearrangement

A radioactive nucleus thus represents the atomic release organisation: it is not simply "more matter", but a bound state whose central saturation produces a directed reorganisation outward. The emitted radiation and decay products are visible consequences of this release.

7.1.2 Quasars as a Galactic Release Class

The quasar is the galactic counterpart of a strongly saturated and actively releasing atomic state. At the centre of a host galaxy, field tension becomes concentrated to an extent that can no longer be carried solely by the galaxy's quiet overall organisation. Release becomes spatially concentrated, coherent, and visible over large distances.

$$Q_{\text{Quasar}} = (\Sigma_G,\, C_{\text{Zentrum}},\, \mathcal{R}_{\text{zentral}},\, \kappa_{G\to 0})\,.$$

Here, Σ_G describes the central galactic saturation, C_{Zentrum} the coherence of the central organisation, $\mathcal{R}_{\text{zentral}}$ the release rate, and $\kappa_{G\to 0}$ the coupling of the emitted release radiation to our observation regime.

The quasar state therefore connects four processes:

central saturation
$\Downarrow$
concentrated release
$\Downarrow$
feedback on gas and outer field
$\Downarrow$
galactic reorganisation

Within this framework, the strong radiation of a quasar is not merely an accompanying feature of the centre. It is the visible release front of a highly saturated central organisation.

7.1.3 Supermassive Galaxies and Heavy Atomic Organisation

A supermassive galaxy is not always a quasar. It can have high central saturation and still remain largely integrated if its release capacity keeps pace with the tension that has built up. This class corresponds to heavy, but stabilised, atomic organisations.

$$\Sigma_G \text{ high}, \qquad \mathcal{E}_{\text{rel},G} \text{ sufficient} \quad \Longrightarrow \quad \text{massive integrated central organisation.}$$

If release capacity is overwhelmed by growing central tension, gas supply, mergers, or a change in outer coupling, the massive integrated galaxy moves into a more active state:

$$\text{supermassive integrated} \longrightarrow \text{centrally activated} \longrightarrow \text{quasar release phase.}$$

The mapping is therefore not

$$\text{quasar} = \text{a specific atom},$$

but

$$\text{quasar} \longleftrightarrow \text{highly saturated atomic release organisation.}$$

7.1.4 Release, Matter Formation, and Feedback

A highly saturated central organisation does not only release its own internal tension. It also changes the conditions in the surrounding field. On the galactic scale, this affects gas, star formation, satellite coupling, and the spatial shape of the outer acceleration profile.

In the Phraqtal Field, release therefore does not form an endpoint, but a transition:

$$\begin{array}{c} \text{saturation} \\ \Downarrow \\ \text{release} \\ \Downarrow \\ \text{outer condensation and reorganisation} \\ \Downarrow \\ \text{new bound structures} \end{array}$$

The atomic counterpart is the reorganisation of an excited or unstable state into new, more stable bound configurations. The galactic counterpart is the effect of a central release phase on gas, stars, and outer partial organisations.

The saturation class therefore complements the previous element–galaxy matrix: helium and noble gases mainly describe closed stability; elements with open outer shells describe coupling and reorganisation; heavy excited or unstable atomic states describe concentrated release. Quasars and active supermassive galaxies belong to this last class.

7.2. Quasars as Cyclical Centres of Release and Matter Formation

In the Phraqtal Field, a quasar is not solely a centre of matter accretion and high-energy radiation. Its central saturation organisation also creates conditions under which new bound matter states can form. The strong luminosity of the quasar is therefore the visible side of a broader release process: field tension stored at the centre is reorganised into radiation, partial organisations, and new bound forms of matter.

The central dynamics have three directions acting at the same time:

$$\text{accretion} \;+\; \text{release} \;+\; \text{matter formation}.$$

Accretion increases central saturation. Release sends part of the bound tension outward as high-energy radiation, outflows, and large-scale field response. Matter formation organises another part of the release dynamics into new stable or pre-stable field modes.

7.2.1 Electron–Positron Pairs as the First Coupling-Capable Form of Matter

In highly saturated release-tension fields, the formation rate of minimally oppositely organised field modes increases within the Phraqtal Field. The first coupling-capable matter organisation consists of electron–positron pairs:

$$\mathcal{F}_{\text{sat}} \longrightarrow e^- + e^+.$$

This pair formation is the first material expression of a release that does not travel outward entirely as freely propagating radiation. Part of the field tension is transferred into two oppositely coupled, locally bound minimal modes.

The pair-formation rate increases with central saturation, local coherence, and available release tension:

$$\dot{N}_{e^-e^+} = \Phi\left(\Sigma_{\text{Zentrum}}, C_{\text{Zentrum}}, \mathcal{R}_{\text{zentral}}\right).$$

The more strongly saturated the centre is and the more coherently its release front remains organised, the stronger the conversion of release tension into minimal coupling-capable forms of matter.

7.2.2 From the Pair State to Bound Matter

Electron and positron do not form the endpoint of matter organisation. In the Phraqtal Field, they are the first coupling-capable stage of a continuing binding sequence:

$$\begin{array}{c}\text{release tension}\\ \Downarrow\\ e^-e^+\\ \Downarrow\\ \text{quark organisation}\\ \Downarrow\\ \text{proton–neutron binding}\\ \Downarrow\\ \text{hydrogen and helium}\end{array}$$

Hydrogen and helium occupy a special position here. They are the first common forms of matter that remain bound over long periods. Hydrogen forms the open, highly coupling-capable initial form. Helium forms the first especially coherent and closed form of material stability.

Matter formation in the quasar environment therefore does not immediately produce complex galactic structures. It begins with minimal coupling states that are gradually transformed into larger bound systems through proton–neutron organisations, gas accumulations, and star formation.

7.2.3 The Cyclical Return

Part of the matter newly formed in the extended central environment does not remain directed outward permanently. Through central binding, inner galactic dynamics, and coupling to gas and star-formation regions, a fraction is returned to the central organisation.

The quasar therefore forms a cyclical return:

$$\begin{aligned}\text{central saturation} &\longrightarrow \text{release radiation} \longrightarrow \text{pair and matter organisation} \\ &\longrightarrow \text{gas and star-formation zones} \longrightarrow \text{feedback to the centre.}\end{aligned}$$

The feedback does not supply the quasar through the unchanged return of the same matter. It brings reorganised matter, energy, angular momentum, and field tension back into the central region in altered form. This creates a continuous, but not static, self-sustaining cycle.

Central saturation accordingly develops from inflow, return flow, release, and material formation:

$$\frac{d\Sigma_{\text{Zentrum}}}{dt} = J_{\text{in}} + J_{\text{rck}} - J_{\text{rel}} - J_{\text{form}}.$$

Here, J_{in} denotes the inflow of matter and tension from outside, J_{rck} the cyclical return from the surrounding reorganisation, J_{rel} the release output, and J_{form} the fraction that passes into new bound matter organisation.

7.2.4 Self-Sustaining Cycle and Limitation

The cyclical self-sustaining process is not unlimited amplification. If every release phase increased the central saturation state more than it reduced it, the system would lose its ordered coherence. The quasar cycle therefore has a natural limitation through outflow, radiation loss, declining feedback efficiency, and the spatial relocation of matter formation.

$$J_{\text{rel}} + J_{\text{form}} \gtrsim J_{\text{in}} + J_{\text{rck}}$$

leads to a weakening of the active quasar phase. If inflow and return flow dominate, central saturation rises again and another active release phase is prepared.

The observable quasar luminosity therefore marks the phase of high central coherence:

$$L_{\text{QSO}} \propto \mathcal{R}_{\text{zentral}}\, C_{\text{Zentrum}}\, \kappa_{G\to 0}.$$

It does not measure the total matter formation of the cycle. It shows the share of the release that is coupled to our regime as high-energy radiation.

7.2.5 Atomic Counterpart of the Quasar Cycle

The atomic counterpart of an active quasar is a strongly excited or unstable central state whose release organisation produces new bound states. In a heavy unstable nucleus, central tension is reduced through decay and particle emission; the decay products can in turn form new bound configurations.

The quasar extends this role to the galactic scale:

$$\text{excited or unstable nucleus} \longleftrightarrow \text{active saturated quasar region.}$$

On both scales, central saturation produces more than radiation. New coupling-capable partial organisations emerge and enter further forms of binding. The crucial difference lies in the scale, duration, and variety of the subsequent reorganisations.

In the Phraqtal Field, the quasar is therefore at the same time a saturation centre, release source, matter-formation zone, and feedback node of the host galaxy.

7.3. Regime-Separated Return between the Quasar Centre and Matter-Formation Zones

The quasar cycle has two different temporal orders. The release front itself moves as a field motion at the invariant speed c. The resulting matter organisation, however, requires the local time of its respective regime: pair formation, binding, gas condensation, star formation, satellite formation, and feedback are step-by-step reorganisation processes.

For a release front across the characteristic distance L,

$$\tau_{\text{Front}} = \frac{L}{c}.$$

The formation of a new bound organisation has its own local reorganisation time

$$\tau_{\text{form}} = \tau\left(\Sigma,\, C,\, \kappa,\, \mathcal{D}\right),$$

where Σ denotes the local saturation, C the coherence, κ the coupling capability, and $\mathcal{D}$ the density of the already existing matter organisation.

The return of gas, stars, satellite material, and bound field structures to the central region again has a galactic integration time:

$$\tau_{\text{rck}} = \tau\left(K_G,\, O_G,\, R_G,\, \mathcal{K}\right).$$

This gives the temporal hierarchy

$$\tau_{\text{Front}} \ll \tau_{\text{form}} \lesssim \tau_{\text{rck}}.$$

The central release front reaches the outer region very quickly as field motion. Visible matter formation follows more slowly because it must first pass through the local formation of coherent partial organisations. The return to the central region is slower than both processes

because it is tied to the large-scale binding and reorganisation of the host galaxy.

7.3.1 The Outer Matter-Formation Zone

The high-energy central release of a quasar acts not only in the immediate core region. It changes conditions in outer gas and field zones. There, the release front can encounter existing condensations, gas flows, and satellite organisations. With sufficient coherence, new bound forms of matter arise.

$$\mathcal{F}_{\text{Zentrum}} \xrightarrow{c} \mathcal{F}_{\text{Auen}} \longrightarrow \mathcal{M}_{\text{neu}}.$$

Outer matter formation therefore does not take place simply as the ejection of already finished matter from the centre. It is a location-dependent reorganisation of the release tension emanating from the centre together with the already existing outer field.

Different developmental stages can overlap in this zone:

$$\begin{array}{c} e^- e^+ \\ \Downarrow \\ \text{bound partial modes} \\ \Downarrow \\ \text{gas organisation} \\ \Downarrow \\ \text{stellar population} \\ \Downarrow \\ \text{satellite or host integration} \end{array}$$

Visible star formation marks an advanced coherence stage. The underlying field and matter-organisation zone already exists before becoming optically visible.

7.3.2 Return to the Central Region

The newly ordered outer matter can take three different paths:

outward integration	:	formation of stable outer stellar and satellite structures,
local binding	:	permanent classification within the disk, gas region, or halo,
central return	:	return of matter and field tension to the quasar centre.

Central return occurs when binding to the host organisation remains stronger than permanent decoupling outward. Gas flows, condensed partial organisations, and integrated satellites then supply matter and reorganisation tension to the central region again.

The quasar does not simply receive back the material it previously ejected. It receives changed feedback:

$$\mathcal{M}_{\text{neu}} + \Delta\mathcal{F}_{\text{Auen}} \longrightarrow J_{\text{rck}}.$$

J_{rck} therefore contains the history of the outer reorganisation. The central organisation is not only strengthened by the return, but also structurally modified.

7.3.3 Phases of the Complete Quasar Cycle

The cyclical process can be divided into six related phases:

I : central saturation through accretion and inner condensation,
II : activation of a concentrated release front,
III : high-energy visibility of the quasar state,
IV : pair formation and matter organisation in outer zones,
V : gas, stellar, and satellite integration,
VI : return of a share of the reorganised matter to the central region.

After phase VI, another saturation phase can begin. The cycle is not periodic in the strict sense, however. Its duration, strength, and spatial form depend on inflow, coherence, Q-activation, outer coupling capability, and the release capacity of the central region.

7.3.4 Observable Sequence within the Phraqtal Field

The separation between regimes results in a time-shifted organisational signature. A central high-energy quasar phase and outer matter formation do not have to reach their maximum optical visibility at the same time. The field front acts first; the visible gas condensation and star formation follow with a local delay; the central return forms the longest-term phase.

$$t_{\mathrm{Quasar}} < t_{\mathrm{uere\ Kohrenz}} < t_{\mathrm{Rckkopplung}}.$$

A host galaxy can therefore carry a past central release phase in the form of younger outer stellar populations, gas structures, satellite couplings, or changed outer organisation, even though the quasar has already weakened at the time of observation.

The quasar cycle thus connects the central region, outer matter formation, and galactic integration into a single, regime-separated organisational dynamic.

8. Empirical Working Status and Testing Architecture

8.1. Current Working Status

The first normalised profile analysis provides several element–galaxy candidates. UGC 01230 has a simple outer organisational signature and lies between a hydrogen-like and a neon-like basic class. NGC 5033 shows a pronounced proximity to several outer atomic states; its transition ratio provisionally places it in the argon-like class. NGC 2955 forms a first candidate of a calcium-like, far-reaching outer class.

The next stage of the analysis determines the strength and radial course of the outer response

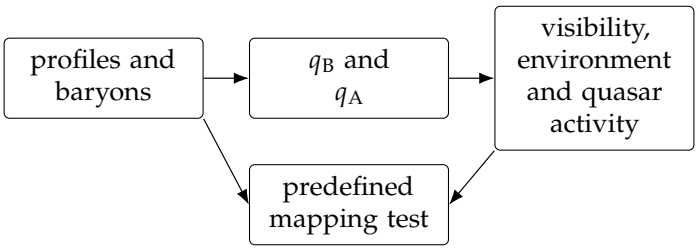


Abbildung 8: Testing architecture: The mapping connects profile shape, activation, baryonic structure, and further observable levels.

for these candidates. Outer shape, transition, response capability, release signature, and temporal organisation are then combined in a common mapping structure.

8.2. Empirical Testing Programme of the Phraqtal Field

The Phraqtal Field does not classify galaxies solely by mass, brightness, or morphology. What matters is their outer coupling organisation, the shape of the acceleration excess, the activation state of the transition zone, the matter-formation structure, and the visibility of their release organisation.

The empirical test is therefore carried out on four levels:

$$\mathcal{P}_{\mathrm{Ph}} = \left(\mathcal{P}_{\mathrm{Profil}}, \mathcal{P}_{\mathrm{Aktivierung}}, \mathcal{P}_{\mathrm{Materie}}, \mathcal{P}_{\mathrm{Sichtbarkeit}}\right).$$

8.2.1 Profile Test: Peak and Diffuse Outer Organisation

The shape axis q_B separates a concentrated Peak organisation from a broadly distributed diffuse outer organisation. This distinction must appear not only in the fitted acceleration profile, but also in the spatial distribution of gas, star formation, and coupled partial organisations.

For Peak systems, the Phraqtal Field expects:

$$q_B \text{ small} \quad \Longrightarrow \quad \text{localised outer reorganisation zone.}$$

This class should more often carry a spatially concentrated gas condensation, a clearly defined transition region, or a local accumulation of outer partial organisations. It is especially relevant for gas-rich, structurally open, and developing galaxies.

For diffuse systems:

$$q_B \text{ large} \quad \Longrightarrow \quad \text{broadly distributed outer field response.}$$

These galaxies should integrate their release and acceleration organisation across a larger radial range. Their outer structure is not necessarily weak; it is less concentrated spatially and more integrated into the host organisation over longer periods.

8.2.2 Activation Test: The Dynamic Switch

The activation axis q_{A} describes whether a dynamic transition scale has formed between the inner and outer regions. The switch is controlled by the transition strength $\mathcal{T}$:

$$q_{\mathrm{A}} = q_{\max}\left[1 + \exp\left(-\frac{\mathcal{T} - \mathcal{T}_{\mathrm{c}}}{w}\right)\right]^{-1}.$$

This gives a directly testable distinction:

$$\begin{aligned} q_{\mathrm{A}} \simeq 0 \quad &: \text{ no strongly pronounced radial switching range,} \\ q_{\mathrm{A}} > 0 \quad &: \text{ developed transition zone between inner and outer organisation.} \end{aligned}$$

Activated systems must show a stronger change in reorganisation behaviour between the inner and outer regions in the radial acceleration profile than non-activated systems. This difference is the central empirical anchor of the Q-switch interpretation.

8.2.3 Matter Test: Open and Closed Families

The element–galaxy matrix leads to a second testable expectation. Open organisational families that correspond in the Phraqtal Field to the hydrogen or re-opened outer-shell class should more often show features of active outer coupling:

$$O_G \text{ high} \quad \Longrightarrow \quad \begin{cases} \text{higher share of gas organisation,} \\ \text{younger visible stellar populations,} \\ \text{stronger satellite or flow coupling,} \\ \text{greater readiness for reorganisation.} \end{cases}$$

Closed families that correspond to the helium or noble-gas organisation should be characterised more strongly by inner integration:

$$C_G \text{ high}, \qquad O_G \text{ limited} \quad \Longrightarrow \quad \text{broadly stabilised overall organisation.}$$

The test does not require individual galaxies to remain permanently in only one class. A galaxy can move between open, integrated, and activated states. What matters is whether the measured profiles and matter indicators carry the organisational state assigned to them.

8.2.4 Visibility Test

The Phraqtal Field separates physical organisation from immediate visibility. Optical brightness alone is therefore not a complete indicator of the strength of an outer field structure.

A coherent partial organisation that is only weakly coupled to our regime can have low optical visibility while remaining dynamically effective:

$$\mathcal{F}_{\mathrm{X}} \neq 0, \qquad \mathcal{V}_{\mathrm{X}\to 0} \ll 1.$$

For the test, this means that optical data must be combined with other carriers of visibility:

$$\mathcal{V}_{\text{gesamt}} = \mathcal{V}_{\text{optisch}} + \mathcal{V}_{\text{Gas}} + \mathcal{V}_{\text{radio}} + \mathcal{V}_{\text{Rontgen}} + \mathcal{V}_{\text{dynamisch}}.$$

A faintly visible dwarf galaxy, a diffuse gas structure, or a satellite stream can therefore be investigated as an incompletely visible partial organisation rather than appearing merely as missing or insignificant matter.

8.2.5 Quasar Test: Time-Shifted Feedback

The quasar cycle requires a temporal sequence of central release, outer matter formation, and feedback:

$$t_{\text{Entlastung}} < t_{\text{uere Materiebildung}} < t_{\text{Rcklauf}}.$$

This gives three distinguishable observational states:

Observational state	Expected organisational signature
Active quasar	strong central high-energy visibility and pronounced release organisation
Fading quasar	declining central emission while outer gas or star-formation structures remain visible
Integrated aftermath	weak or absent quasar visibility with continuing traces of earlier outer reorganisation and feedback

The empirical focus is therefore not on a single snapshot. It is on the joint investigation of central emission, gas distribution, stellar populations, satellite coupling, and the radial acceleration profile.

8.2.6 Core Predictions

The strongest testable statements of the Phraqtal Field can be summarised as follows:

P1 : The bimodality of the outer field shape remains after controlling for basic baryonic quantities.

P2 : The Q-switch separates activated from non-activated radial transition organisations.

P3 : Peak, transition, and diffuse profiles differ systematically in gas, coupling, and visibility structure.

P4 : Quasar activity, outer matter formation, and feedback show a temporally staged organisational sequence.

P5 : Weakly visible outer partial organisations can remain dynamically relevant when their regime coupling is maintained.

These predictions make the Phraqtal Field empirically falsifiable as an organisational framework and allow its element–galaxy mapping to be tested step by step against independent data.

8.3. Operational Testing Architecture for the Element–Galaxy Mapping

The empirical test of the Phraqtal Field must separate the organisational mapping from subsequent interpretation. The classes, variables, and analysis sequence are therefore specified before the respective comparison analysis.

The testing architecture combines four data levels:

$$\mathcal{A}_{\mathrm{Ph}} = \left(\mathcal{A}_{\mathrm{Dyn}}, \mathcal{A}_{\mathrm{Bar}}, \mathcal{A}_{\mathrm{Vis}}, \mathcal{A}_{\mathrm{Env}}\right).$$

8.3.1 Dynamic Level

The dynamic level contains the radial acceleration organisation of a galaxy. It starts with the observed acceleration $g_{\mathrm{obs}}(r)$ and the baryonically reconstructed acceleration $g_{\mathrm{bar}}(r)$:

$$\Delta g(r) = g_{\mathrm{obs}}(r) - g_{\mathrm{bar}}(r).$$

The positive excess structure $\Delta g(r)$ is not read as an independent additional matter component, but as a regime-effective response of the baryonically bound field organisation. For each galaxy, four dynamic quantities in particular are recorded:

$$\left(q_{\mathrm{B}}, q_{\mathrm{A}}, \mathcal{T}, R_{\mathrm{tr}}\right).$$

Here, q_{B} describes the shape class of the profile, q_{A} the activation state, $\mathcal{T}$ the strength of the inner–outer transition, and R_{tr} the characteristic transition region.

8.3.2 Baryonic Level

The baryonic level describes which bound matter organisation is connected with the dynamic field response:

$$\mathcal{A}_{\mathrm{Bar}} = \left(M_{\star}, M_{\mathrm{Gas}}, f_{\mathrm{Gas}}, \Sigma_{\star}(r), \Sigma_{\mathrm{Gas}}(r)\right).$$

This includes stellar mass, gas mass, gas fraction, and radial stellar and gas profiles. These quantities alone do not determine the assignment to an element family. However, they show whether a galaxy is characterised more strongly by open outer coupling, closed integration, or active reorganisation.

A hydrogen-like open organisational family is investigated in particular through high gas involvement, visible outer condensation, and reorganisation capability. A helium-like or noble-gas-like closed family is investigated through high inner stabilisation, limited outer openness, and a broadly integrated field response.

8.3.3 Visibility and Coherence Level

The visibility level contains more than optical brightness. It captures different carriers of a visible or indirectly visible field organisation:

$$\mathcal{A}_{\mathrm{Vis}} = \left(L_{\mathrm{opt}}, L_{\mathrm{IR}}, L_{\mathrm{Radio}}, L_{\mathrm{X}}, \mathrm{SFR}, \mathcal{S}_{\mathrm{Spektrum}}\right).$$

L_{opt}, L_{IR}, L_{Radio}, and L_{X} denote the respective observable radiative power, SFR the star-formation rate, and $\mathcal{S}_{\mathrm{Spektrum}}$ the spectral state information.

This level tests the central statement that visibility is a coupling state. A galaxy with low optical visibility can be highly organised dynamically and materially if its field organisation becomes readable in other wavelength ranges or through its acceleration structure.

8.3.4 Environmental and Coupling Level

The environmental data capture outer partial organisations:

$$\mathcal{A}_{\mathrm{Env}} = \left(N_{\mathrm{sat}}, M_{\mathrm{sat}}, \mathcal{S}_{\mathrm{Strom}}, \mathcal{G}_{\mathrm{Gruppe}}, \mathcal{K}_{\mathrm{uer}}\right).$$

This includes satellite number, satellite mass, tidal or material streams, group membership, and the reconstructable outer coupling structure $\mathcal{K}_{\mathrm{uer}}$.

This level is decisive for the electron–satellite analogy. The relevant quantity is not simply the number of observed companions, but their effective coupling, coherence, visibility state, and degree of integration into the host organisation.

8.3.5 Predefined Classification

Each galaxy is first assigned to a dynamic shape and activation class independently of subsequent group comparisons:

$$C_G = (C_{\text{Form}}, C_{\text{Akt}}) .$$

The shape class is:

$$C_{\text{Form}} = \begin{cases} \text{Peak}, & q_{\text{B}} < 1.2, \\ \text{Transition}, & 1.2 \le q_{\text{B}} < 2.5, \\ \text{Diffuse}, & q_{\text{B}} \ge 2.5. \end{cases}$$

The activation class is:

$$C_{\text{Akt}} = \begin{cases} \text{not activated}, & \mathcal{T} < \mathcal{T}_{\text{c}}, \\ \text{switching range}, & \mathcal{T} \simeq \mathcal{T}_{\text{c}}, \\ \text{activated}, & \mathcal{T} > \mathcal{T}_{\text{c}}. \end{cases}$$

Only after this classification are gas organisation, visibility, star formation, satellite coupling, and environmental structure compared.

8.3.6 Mapping Protocol

The element–galaxy mapping is carried out in four steps:

$$\begin{aligned} \text{I} &: \text{determine dynamic shape and activation,} \\ \text{II} &: \text{record inner and outer baryonic organisation,} \\ \text{III} &: \text{combine visibility and coherence carriers,} \\ \text{IV} &: \text{classify satellite and environmental coupling.} \end{aligned}$$

The result is not a rigid naming of a galaxy after a single chemical element. It is a multidimensional organisational mapping:

$$\mathcal{Z}_G = \left(K_G,\, O_G,\, R_G,\, C_G,\, q_{\text{B}},\, q_{\text{A}}\right) .$$

This state description can then be compared with the atomic organisational families hydrogen, helium, open outer shells, closed multi-shell structures, and saturation or release states.

8.3.7 Reproducibility

Every later analysis must document the input files used, variable definitions, quality filters, classification limits, sample sizes, and analysis rules. The Phraqtal Field interpretation is thereby not tied to individual post hoc examples, but to reproducible comparisons between predefined organisational classes.

9. The Visible Regime Universe in Subregime Compression

In the expanding standard model, the present-day diameter of the observable universe is approximately 93 billion light-years [4, 5]. In the static Phraqtal Field, the visible size follows directly from the light-travel distance. Since every tension front moves at the invariant speed c, a visible range of 13.8 billion years corresponds to a radius of 13.8 billion light-years:

$$R_{\text{vis}} = c \cdot 13{,}8\,\text{Gyr} = 13{,}8\,\text{Gly} \simeq 1{,}306 \times 10^{26}\,\text{m}. \tag{87}$$

The visible regime universe therefore has a diameter of

$$D_{\text{vis}} = 2R_{\text{vis}} = 27{,}6\,\text{Gly} \simeq 2{,}611 \times 10^{26}\,\text{m}. \tag{88}$$

The scale nesting recognisable in the Phraqtal Field connects the outer atomic organisation with the outer galactic organisation through a length corridor of

$$\mathcal{L}_{E,G} \simeq 3{,}0 \times 10^{30} \text{ to } 1{,}2 \times 10^{31} \qquad \text{(dimensionless size ratio)}. \tag{89}$$

If the same order of magnitude is transferred to the nesting of the visible regime-universe into a subregime, this gives

$$D_{\text{sub}} = \frac{D_{\text{vis}}}{\mathcal{L}_{E,G}} \simeq 2{,}2 \times 10^{-5} \text{ to } 8{,}7 \times 10^{-5}\,\text{m}. \tag{90}$$

This corresponds to a diameter of

$$D_{\text{sub}} \simeq 0{,}022 \text{ to } 0{,}087\,\text{mm} = 0{,}0022 \text{ to } 0{,}0087\,\text{cm}. \tag{91}$$

The visible regime-universe therefore appears smaller than 0,1 mm in the subregime. Along a distance of one centimetre, this compressed regime size could be arranged more than one hundred times in succession. The extraordinarily small representation size is a direct consequence of recursive scale nesting: a regime is not separated from its subregime, but is contained within its finer resolution as a large-scale field organisation.

9.1. Energetic and spatial size ratios in the Phraqtal Field

In the previous work *Gravitational Light* [9], the ratio

$$\Lambda_n = \frac{E^{(n)}_{\text{max}}}{E^{(n)}_{\text{min}}} \approx \frac{3M_\odot}{m_e} \approx 6{,}5 \times 10^{60} \tag{92}$$

describes the energetic range of a complete regime. It extends from the smallest stable, independently coupling field organisation to the highest locally attainable saturation state, approximated here by a minimal black-hole mass. This ratio therefore does not initially describe a spatial distance, but rather the difference between minimal coupling capacity and maximal field compression. The spatial size transfer between atomic and galactic

organisational forms is described independently by the length corridor

$$\mathcal{L}_{E,G} \simeq 3{,}0 \times 10^{30} \text{ to } 1{,}2 \times 10^{31} \qquad \text{(dimensionless size ratio).} \tag{93}$$

The fact that both orders of magnitude are close to one another is remarkable because the square of the spatial factor also approaches 10^{61}:

$$\left(10^{30{,}5}\right)^2 \approx 10^{61}. \tag{94}$$

Thus, two interconnected but different aspects of the same regime organisation become apparent: Λ_n describes the energetic depth of the field organisation, while $\mathcal{L}_{E,G}$ describes its spatial extent across different scales.

10. Consciousness, memory and sensation as an outlook

The Phraqtal Field describes observation as coupling between an incoming field modulation and a bound observer organisation. The observer is therefore itself part of the physical organisational chain:

field motion
⇓
bound matter
⇓
biological organisation
⇓
observation

This statement is sufficient to place perception within a physical framework. It does not, however, yet provide a complete description of memory, self-reference or consciousness. These arise only when a biological field organisation not only responds to a modulation, but also stabilises its own states over time, compares them with one another and incorporates them again into its further reorganisation.

10.0.1 Memory as a stable trace of reorganisation

In the Phraqtal Field, memory can be understood as a permanently modifiable binding structure. An incoming modulation leaves not only a momentary signal, but changes the probability of later couplings:

$$\mathcal{M}_{t+\Delta t} = \mathcal{M}_t + \Delta\mathcal{K}_t.$$

Here, $\mathcal{M}_t$ does not denote an abstract information-storage object outside matter, but rather the respective state of a bound and reactivatable organisational structure. $\Delta\mathcal{K}_t$ is the change in its future coupling capacity produced by experience, response and feedback.

A memory state is therefore a stabilised trace of reorganisation:

perception
⇓
reorganisation
⇓
stable trace
⇓
altered later perception

10.0.2 Sensation as integrated inner readability

Sensation does not arise from a single external stimulus. It arises from the integration of many field modulations coupled simultaneously and with temporal staggering within a highly organised biological binding form.

$$\mathcal{E}_{\text{empf}} = \mathcal{I}\left(\mathcal{V}_1, \mathcal{V}_2, \ldots, \mathcal{V}_n; \mathcal{M}; C_O\right).$$

Here, $\mathcal{I}$ denotes inner integration, $\mathcal{V}_i$ individual coupled perceptual modulations, $\mathcal{M}$ the existing memory organisation and C_O the coherence of the observer.

The depth of sensation in an organism therefore depends not only on the strength of a stimulus. It depends on how many layers of modulation are distinguished, integrated and connected with previous inner states:

$$\text{sensation depth} \propto \text{coupling diversity} \times \text{memory integration} \times \text{inner coherence}.$$

10.0.3 Consciousness as recursive self-coupling

Within the Phraqtal Field, consciousness begins where a bound organisation not only couples to its environment, but also includes its own ongoing coupling and memory states in its further organisation:

$$\begin{aligned}&\text{environmental coupling} + \text{memory coupling} + \text{self-coupling}\\ &\qquad\longrightarrow\ \text{recursive observer organisation}.\end{aligned}$$

The central role here is played by recursion. A conscious observer does not process only a present field modulation. It also processes its own position in relation to this modulation, its already existing memory traces and the expected consequences of a possible response.

$$O_{t+1} = \mathcal{F}\left(\mathcal{U}_t, \mathcal{M}_t, O_t\right).$$

$\mathcal{U}_t$ denotes the environmental coupling, $\mathcal{M}_t$ the memory organisation and O_t the current inner observer state.

10.0.4 Position within the Phraqtal Field

Consciousness does not form a second reality separated from matter and the field within the Phraqtal Field. It is the highly integrated inner readability of a biological field organisation.

Its physical basis lies in bound matter, stable traces of reorganisation, coherence and recursive self-coupling.

The present work follows this line only as far as its basic organisational form. A separate study would have to develop the connection between neuronal binding organisation, temporal integration, memory formation, self-modelling and subjective sensation in a targeted manner and test it against biological data.

10.1. Memory, depth of sensation and consciousness

Memory arises in the Phraqtal Field when a reorganisation does not return completely to its initial state. A part of the coupled change remains as a stabilised trace within the bound organisation. This trace changes the conditions under which later field modulations are received, weighted and answered.

event
⇓
reorganisation
⇓
stabilised trace
⇓
altered later response

At the material level, this principle appears as a persistent structure, binding state or change of state. In biological organisations, the trace is integrated by highly coupled systems with repeated feedback. Memory is therefore not information existing alongside matter, but the continuing organisational history of material field bindings.

The depth of sensation of an organisation depends on how many levels of an incoming field change it can receive, distinguish, connect with one another and include in its own feedback at the same time. A simple organisation responds immediately. A deeply integrated organisation, by contrast, can connect present stimuli with stored patterns, inner states, expectation structures and higher-level goals.

As a working concept, the depth of sensation can be written as

$$\mathcal{E}_{\text{tief}} = \mathcal{F}\,(C, R, G, I)\,,$$

where C denotes the coherence of the partial organisations, R the depth of feedback, G the stably stored organisational history and I the integration capacity of the overall organisation. A high depth of sensation does not require merely many individual processes. It requires these processes to form a coherent inner readability.

Within the Phraqtal Field, consciousness denotes the recursive self-coupling of an organisation that not only processes its own states, but also continually refers them back to itself within an ongoing, integrated order of the present. The organisation thereby becomes readable to itself in its own change. Perception, memory, expectation and action do not form separate stages, but different directions of the same feedback.

$$\text{perception} \leftrightarrow \text{memory} \leftrightarrow \text{self-reference} \leftrightarrow \text{action}.$$

Consciousness therefore does not stand outside the physical world. It is a particularly deep form of materially bound, coherent and recursively integrated field organisation. Its distinctive feature does not consist in leaving the field order, but in this order developing within it an ongoing inner view of its own states.

10.2. Overview of terms and symbols

This directory brings together the basic terms and working quantities used in the Phraqtal Field. It serves the consistent readability of the atomic, galactic and observation-related sections.

Tabelle 17: Central terms of the Phraqtal Field.

Term	Meaning
Qu-Foundation Q	not directly measurable fundamental level of the UQSH, from which physically readable field organisation emerges
Field form Φ	physically real, regime-dependent readable form of the Qu-Foundation
$B(\Phi)$	moving field form capable of carrying tension and organisation; basis of free radiation and bound matter
Phraqtal Field	designation for the physically holographic, scale-relative organisational mode of the Phraqtal Field
Regime	independent range of resolution, time order, coupling capacity, stability and visibility within the same field medium
Subregime / Superregime	organisational and coupling levels located relatively below or above a regime
Coupling filter	regime-dependent condition determining which field organisation becomes measurable or visible
Release	reorganisation of a saturated or highly stressed field organisation into radiation, new binding, outflow or feedback
Saturation	state in which additional local densification can no longer be carried as further central deepening and a reorganisation begins

The central state description of a bound organisation is therefore:

$$\mathfrak{P}_X = \left(K_X,\, O_X,\, \Sigma_X,\, C_X,\, R_X,\, q_{\mathrm{B},X},\, q_{\mathrm{A},X},\, \mathcal{V}_X\right).$$

It makes it possible to describe atomic systems, galaxies, quasars and observer organisations in a common, scale-relative language.

Tabelle 18: Organisational variables for a system X.

Symbol	Designation	Meaning
K_X	Binding strength	central stabilisation and ability to carry outer states in an ordered manner
O_X	Opening	outer coupling and reorganisation capacity
Σ_X	Saturation state	ratio of central field tension to the available release capacity
C_X	Coherence	inner ordered stability of a bound or propagating field organisation
R_X	Readiness for reorganisation	tendency of a system to transfer its existing binding form into a new state
$\mathcal{V}_X$	Visibility	strength with which an organisation couples to an observation regime
$\kappa_{X\to Y}$	Coupling strength	effectiveness of the connection between two organisations or regimes
$\mathcal{R}_X$	Release rate	rate at which a system transfers field tension into reorganisation or radiation

Tabelle 19: Galactic differentiation quantities.

Symbol	Designation	Meaning
q_{B}	Bimodality axis	spatial form of the outer field and acceleration organisation
q_{A}	Activation axis	dynamic state of the inner–outer transition scale
$\mathcal{T}$	Transition strength	measure of the change between inner and outer reorganisation behaviour
R_{tr}	Transition radius	characteristic radial zone of the field switch
$\Delta g(r)$	Acceleration excess	difference between observed and baryonically reconstructed acceleration
J_{in}	Inflow	matter and field tension entering the central region
J_{rel}	Release output	centrally emitted radiation, outflow and reorganisation power
J_{rck}	Feedback	share of reorganised outer matter and field tension that acts back on the host organisation

11. Standard Model and Phraqtal Field in comparison

The Phraqtal Field denotes the physical fundamental reality from which all physical states arise. Matter, radiation, gravitation, energy, heat, pressure, rigidity, elasticity, motion, saturation and release are, in this interpretation, not separate fundamental substances, but different organisational forms and states of the same physically effective field medium.

The Standard Model of cosmology classifies the observed contributions of the universe into separate categories. These include baryonic matter, dark matter and dark energy. The proportions of approximately 4,9% baryonic matter, 26,8% dark matter and 68,3% dark energy shown in the left-hand diagram correspond to the rounded cosmological density fractions of the Planck-ΛCDM analysis [4].

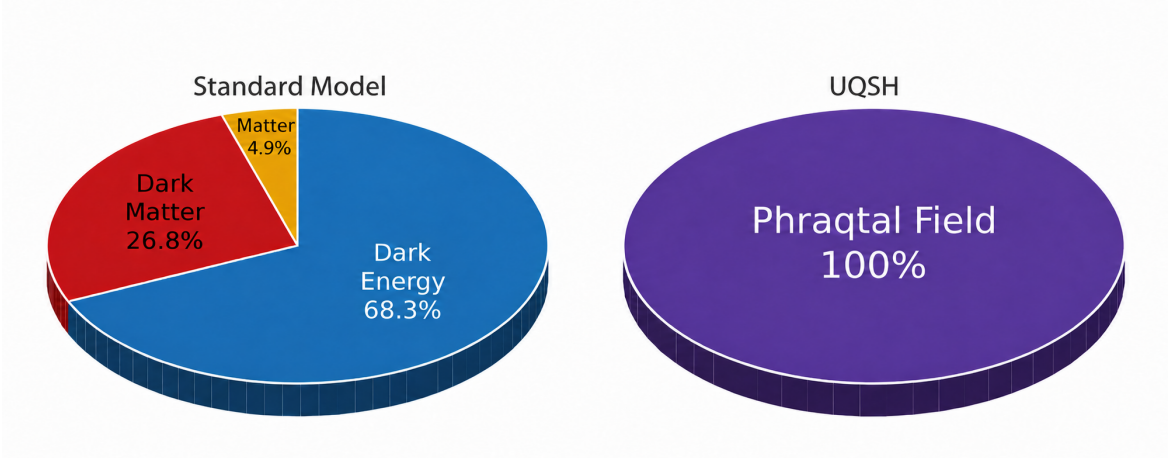


Abbildung 9: Schematic comparison of the cosmological energy fractions in the Standard Model with the unified organisational description of the Phraqtal Field. The Standard Model distinguishes visible matter, dark matter and dark energy as separate fractions. In the Phraqtal Field interpretation, these phenomena are described as different organisational and coupling states of a common field medium.

Figure 9 presents the fundamental difference between the two modes of description. The left-hand diagram shows the Standard Model-based accounting of the cosmological contributions. There, the three fractions are identified as separate contributions to the total energy density of the universe.

The right-hand diagram summarises the Phraqtal Field interpretation. Matter, dark matter and dark energy are not assumed here to be three independent fundamental components. They are different physical states, coupling forms and modes of reorganisation of a common field medium. Their different properties arise from variations in binding, density, coherence, mode of motion, saturation, coupling capacity and regime scale.

The unified description does not mean that these states are identical. Water is not the same as air, gravitation is not the same as radiation and matter is not the same as heat. However, all these phenomena consist of the same physical fundamental reality and differ through their respective field organisation.

The Standard Model therefore primarily describes the observed division into different

cosmological contributions, while the Phraqtal Field denotes the common physical basis of these contributions. The mathematical description classifies the measurable states of the Phraqtal Field and makes their properties comparable at different grid sizes and regime scales. The Phraqtal Field itself is not a mathematical quantity adjusted retrospectively, but the physical unity from which the observable states arise.

11.1. Conclusion

The Phraqtal Field formulates a unified organisational framework for material binding, visibility, saturation, release and reorganisation. Its starting point is the Qu-Foundation Q, whose physically readable field form Φ acts in different regimes as radiation, matter, gravitation, time order and observation.

The central statement of the framework is:

$$\text{matter} = \text{bound coherent field motion},$$

while radiation describes the freely propagating form of local field reorganisation. Visibility arises when a field modulation is connected to the coupling filter of an observer or measurement regime. What is not visible is therefore not necessarily physically unorganised; it may lie below, above or outside the immediate coupling capacity of our regime.

The atom–galaxy correspondence is described in the Phraqtal Field through common organisational roles:

$$\{\text{centre}, \text{binding}, \text{outer occupancy}, \text{saturation}, \text{release}, \text{reintegration}\}.$$

Atomic nuclei and galactic central regions each carry the inner binding of their overall organisation. Electrons assume the role of bound outer coupling modes at the atomic scale; gas, stellar populations, satellites and outer field structures assume the corresponding role at the galactic scale. Hydrogen represents open initial organisation, helium the first stable closed form of matter, and later element families different ratios of inner binding and outer reorganisation capacity.

The bimodality of galactic acceleration profiles and the dynamic Q-switch provide two distinguishable organisational axes:

$$(q_{\mathrm{B}}, q_{\mathrm{A}}) = (\text{spatial form}, \text{dynamic activation}).$$

They allow a differentiated classification between concentrated peak organisations, diffuse outer organisations, non-activated states and activated transition zones. The assignment to element families is therefore not based on external similarity, but on the position of a galaxy in the common space of binding, opening, coherence, saturation and reorganisation.

Within this framework, quasars form highly saturated release centres. Their strong luminosity shows a concentrated coupling of central saturation, coherent release and regime-effective radiation:

saturation
⇓
release
⇓
matter formation
⇓
feedback

The matter-forming side of the quasar cycle begins with minimal coupling-capable field organisations and proceeds through further bound stages to gas, stellar and satellite structures. Some of this reorganised matter can be integrated into the host organisation or returned to the central region. The quasar is therefore at once a saturation centre, release source, matter-formation zone and feedback node.

The empirical core of the Phraqtal Field lies in the joint testing of dynamic profiles, baryonic organisation, visibility and outer coupling. The key research questions are:

1. Do peak and diffuse outer organisations remain distinct profile families?
2. Does the Q-switch separate activated and non-activated transition organisations?
3. Do these classes differ systematically in
   gas, star formation, visibility and satellite coupling?
4. Do quasar hosts show the expected temporally staggered sequence of
   central release, outer reorganisation and feedback?

The Phraqtal Field concludes with the statement that physical reality is not built from separate substances and forces. It consists of a continuous Phraqtal Field that binds, moves, releases, becomes visible and produces complex observer organisations in different regimes.

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Author contribution

Ukshin Q. Rexhepi developed the theoretical framework, formulated the research questions, carried out the conceptual analysis and is responsible for the interpretation of the results and the final version of the manuscript.

Use of computational tools

Computational tools were used for numerical analyses, code development, statistical tests, visualisation and language editing. The scientific questions, the choice of testing architecture, the contentual evaluation of the results and responsibility for all statements lie with the author.

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